



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY
DEPARTMENT OF MECHANICAL ENGINEERING.

SECOND SEMESTER EXAMINATION 2016/2017 ACADEMIC SESSION

Course Title: WORKSHOP PRACTICE (MEE 102)

Time Allowed: 20 Minutes

Instruction(s): *There are six sections in all. Answer 7 questions each from sections A, B, C, D & E, and 5 questions from Section F, making 40 questions in all.*

CANDIDATE'S NAME: **MATRIC NO.**.....

SCHL: **DEPT:**

DATE:

SECTION A: FITTING

1. The assembly of components parts of an equipment or a machine is called.....
 - A. Fitting +++
 - B. Bench work
 - C. Fixing
 - D. Sheet metal work
2. Sheet metal can be cut using the following tools EXCEPT.....
 - A. Snips
 - B. Vice +++
 - C. Hacksaw
 - D. Guillotine
3. Hacksaw blade is made of.....
 - A. Carbon or High Speed Steel +++
 - B. Hardened Steel
 - C. Forged Steel
 - D. High Carbon Steel
4. Chisels are made of

- A. Hardened Steel
- B. High Speed Steel
- C. Forged High Carbon Steel +++
- D. High Grade Steel

5. Taps are used for making external thread

- A. True
- B. A&B
- C. False +++
- D. None of the above

6. Which of the following is used to determine the type of hammer?

- A. Point
- B. Wedges
- C. Pein +++
- D. Eye

7. Files are made of.....

- A. High Speed Steel
- B. Hardened Stainless Steel
- C. Hardened Mild Steel
- D. Hardened High Grade Steel +++

8. Rivets are not permanent mechanical fasteners.

- A. None of the above +++
- B. False
- C. True
- D. All of the above.

9. Anvil is made of

- A. Forged steel
- B. Hardened Mild Steel +++
- C. High Speed steel
- D. Mild Steel

10. Which of the following is not a typical part of a file.....

- A. Point

- B. Shoulder
- C. Eye +++
- D. Face

11. The following are marking out tools EXCEPT.....

- A. Punch
- B. Scriber
- C. Try Square
- D. File +++

12. Which of the following is not a method of filling.....

- A. Draw filling
- B. Finish filling
- C. Start filling +++
- D. Cross filling

13. Heavy duty type of power driven guillotine can be used to cut 12 mm steel plate

- A. True +++
- B. False
- C. All of the above
- D. None of the above

14. When production is carried out with the use of machine tools, it is referred to as

- A. Machine tools
- B. Lathe cutting
- C. Bench work +++
- D. None of the above

15. The drilled hole in riveting operation must not be larger than the rivet to allow the rivet expand when formed.

- A. True
- B. False +++
- C. All of the above
- D. None of the above

16. The depth of a blind hole can be accurately determined by-----

- A. Vernier Caliper +++
- B. Measuring tape
- C. Steel rule
- D. Micrometer screw gauge

17. The following information's can be obtained on sequence of operation EXCEPT?

- A. Tools used
- B. Operation performed
- C. Materials of those parts
- D. The risk involved in each operation +++

18. To produce a bolt on the fitters bench you need-----

- A. Die +++
- B. Surface plate
- C. Micrometer screw gauge
- D. Tap

19. The hammer used for light work to avoid damage to the surface of such component is called?

- A. Soft hammer +++
- B. Sledge hammer
- C. Straight pen
- D. Ball Pein hammer

20. The screw of a fitter's bench vice is made of----- thread

- A. Square or buttress threads +++
- B. Ball pein
- C. Die
- D. Tap

21. An operation whereby a cylindrical pin with head is used to fastening two or more plates together is called-----

- A. Welding operation
- B. Cross cutting operation
- C. Riveting operation +++
- D. Chipping and chiseling operation

22. Which of the following is a measuring tool?

- A. Micrometer screw gauge
- B. Engineer Square +++
- C. Steel rule
- D. Electrode holder

23. The smaller hole found on the top surface of an anvil is called

- A. Pinch hole
- B. Process hole
- C. Punch hole +++
- D. Flat surface

24. To effectively carry out chiseling and chipping operation by a fitter you need chisel and

- A. Snip
- B. Mallet
- C. Die stock
- D. Hammer +++

25. The parts of a hacksaw which enable the tensioning of the blade is called

- A. Wing Nut
- B. But and Square nut +++
- C. Square thread
- D. Pin

26. What is the effect of pinning in filing operation?

- A. Lively easy cut
- B. Loss of cutting efficiency of the file
- C. Blow hole
- D. Dead smooth +++

27. To effectively hold a tool for cutting internal thread, you need

- A. Tap Wrench +++
- B. Pliers
- C. Die stock
- D. Ring spanner

28. The followings are bench work and fitting auxiliary tools EXCEPT

- A. Tee- Block
- B. Set of spanner
- C. Wire brush
- D. Vee- block +++

29. is a hard fibrous tissue found in many plants

- A. Wood +++
- B. Fibre cellulose
- C. Cell walls
- D. Cellulose

30. Why it is that caliper cannot be used for direct measurement?

- A. Too small for the purpose
- B. Too large for the purpose
- C. It is not calibrated +++
- D. For accuracy purpose

31. Which of the followings is a work holding device?

- A. Anvil
- B. Bench Vice +++
- C. Screw driver
- D. Fitters bench

32. Is generally obtained from a tree with deciduous or broad leaves

- A. Shisham
- B. Hardwood +++
- C. Softwood
- D. Bonded wood

33. is used for cutting sheet metals of various nature and thickness

- A. Cutting blade
- B. Snip +++
- C. Anvil
- D. Tube cutter

34. The type of vice for holding tapered work piece is called vice.

- A. Swift Rig
- B. Tap Wrench
- C. Swivel +++
- D. Tail stock

35. According to roughnessfile is excellent for hard metals.

- A. First cut
- B. All
- C. First and second cut
- D. Second cut +++

36. Which of the following is not a type of file according to roughness?

- A. Straight +++
- B. Smooth
- C. Bastard
- D. Second cut

37. The blade of a hacksaw can be made of

- A. High carbon steel
- B. Nickel-alloy steel
- C. carbon steel +++
- D. forged steel

38. Which of the following is not a part of a hacksaw?

- A. Frame
- B. Peg +++
- C. Wing nut
- D. Handle

39. One these is an example of Pin punches EXCEPT

- A. Tapered +++
- B. Dot
- C. Centre
- D. Tap

40. Conversion of tree into a timber or wood log is called

- A. Seasoning
- B. Salten
- C. Felling +++
- D. Cracking

41. Which of the following can be the length of a file used for light work?

- A. 180 mm +++
- B. 210 mm
- C. 260 mm
- D. 300 mm

42. Hammer can be used for the following operations except

- A. Punching
- B. Chipping
- C. Riveting
- D. Sawing +++

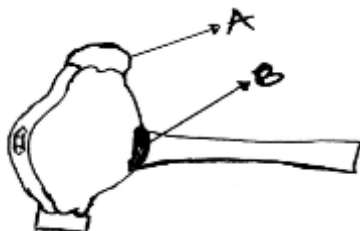
43. The part of the hacksaw which performs the function of holding hacksaw blade in position is called ...

- A. Wing Nut
- B. Pins/Pegs +++
- C. Tension Saw
- D. Bolt and Nut

44. The clogging of file teeth with small particles of metal there by reducing the cutting efficiency of the file is called

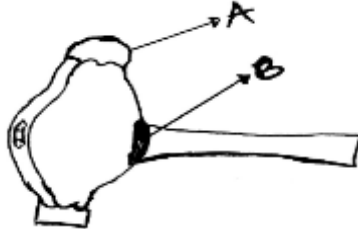
- A. Pin Hole
- B. Pinning +++
- C. Pinn
- D. Plumb line

45. The part labeled A in the figure below is called



- A. Pain
- B. Pin Head
- C. Eye
- D. Pein +++

46. The part labeled B in the figure below is called.....



- A. Pein
- B. Eye +++
- C. Pein Eye
- D. Head

47. is used for bending small rods that are round to form rigs, striking metal along its length.

- A. Straight pein
- B. sledge hammer
- C. Ball pein +++
- D. None of the above.

48. All these are part of hammer EXCEPT

- A. Head Stock +++
- B. Head
- C. Face
- D. Pein

49. is the removal of thick layers of metal by means of cold chisels.

- A. Strips
- B. Chipping +++
- C. Flattening
- D. Anvil

50. The following tools are used during chiseling operation EXCEPT

- A. Chisels
- B. Hammer
- C. Dies +++
- D. Anvils

51. The following are type of punches EXCEPT

- A. Centre punch
- B. Straight punches +++
- C. Pen punches
- D. Dot punches

52. The body of a vice is fitted with Steel jaws

- A. All of the above
- B. Semi-detachable
- C. Non detachable
- D. Detachable +++

53. The following are type of vices EXCEPT

- A. Optimum vice +++
- B. Bench vice
- C. Pin vice
- D. Swivel vice

54. One of these is not an example of chisel.

- A. Quarter round chisel
- B. Diamond point chisels
- C. Cross cut chisel
- D. Round chisel +++

55. The following are example of snips EXCEPT

- A. Cross-cut snips +++
- B. Straight snips
- C. Curved snips
- D. Universal pattern snips

56. All these are part of a typical hacksaw EXCEPT

- A. Hack saw blade

- B. Wing nut
- C. Hardie hole +++
- D. Tension screw

57. A hardened piece of high grade steel with slanting rows of teeth used to cut smooth or fit metal parts.

- A. File +++
- B. Hacksaw
- C. Snips
- D. Dies

58. The type of tap that is used for starting the thread is called

- A. Plug tap
- B. Taper tap +++
- C. Taper dies
- D. All of the above

59. is a finishing process that can obtain a degree of accuracy much higher than that obtained with chisel or file.

- A. Welding
- B. Brazing
- C. Scrapping +++
- D. Riveting

60. The following are types of cleaning tools EXCEPT

- A. Wire guzzle +++
- B. Flexible scrapper
- C. Wire brush
- D. Bristle head

61. The type of solid dies for rectifying damage to existing threads or running an existing thread down to size is called

- A. Die nuts +++
- B. Die heads
- C. Circular dies
- D. Rectangular dies

62. The following are types of files EXCEPT.

- A. Triangular file
- B. Half -round file
- C. Semi-round file +++
- D. Knife-edge file

63. is the clogging of the file teeth with particles of metal causing the file to its cutting efficiency.

- A. Pinning +++
- B. Filling
- C. Sizing
- D. Grinding

64. Which of the following tools does not belong in the group

- A. Hammer
- B. Scraper
- C. Lathe +++
- D. Hacksaw

65. The accuracy of the pedestal grinding machine is

- A. 0.2 mm
- B. 0.025 mm +++
- C. 0.0025 mm
- D. 0.52 mm

66. When ordering a file, you must know it's; I. length II. Degree of roughness III. Shape. IV. Type of cut

- A. I only
- B. I and II only
- C. I, III and IV only
- D. All of the above +++

67. Removal of clogged particles from the teeth of a file is called.....

- A. Pinning +++
- B. Peening
- C. Deluging

D. Scraping

68. Safety goggle, overall, safety boots, wire harness and helmets are collectively called

A. PEP

B. EPP

C. PPE +++

D. EPE

69. Two sheet metals are 6mm thick and 7mm thick and are required to be joined by rivets.

The diameter of rivet used should be

A. Three times the thickness of the thicker sheet metal +++

B. Two times the thickness of the thicker sheet metal

C. Sum of the thickness of both sheet metal

D. Five times the thickness of the less thick metal.

70. Which of the following does not belong in the group?

A. Center punch

B. Template

C. Snips +++

D. Scriber

71. Which of the following is not a sheet metal operation?

A. Cutting

B. Stream cutting +++

C. Rolling

D. Bending

72. The scribing of lines on a work piece to indicate the area to be cut is known as

A. Marking off +++

B. Marking in

C. Marking on

D. Marking to

73. is a hole on the anvil used in providing clearances

A. Screw Hole

B. Drill hole

- C. Bore hole
- D. Punch Hole +++

74. The size of a typical hack saw can be determined by

- A. Measuring the height of the frame
- B. Measuring the distance between the pegs +++
- C. Measuring the distance between legs
- D. between the wing-nut and the handle

75. A single cut file has teeth serrated at

- A. 70°
- B. 50°
- C. 60° +++
- D. 80°

76. Spanners, pliers and screw drivers are collectively called

- A. Auxiliary tools +++
- B. Ancillary tools
- C. Peculiar tools
- D. Domestic tools

77. In the engineering workshop, a work piece that has not been worked on is called

- A. Plank
- B. Blank +++
- C. Tank
- D. Rank

78. Production of articles in the workshop using only hand tools is known as

- A. Workbench
- B. Fitting
- C. Fixing
- D. Bench work +++

79. The part of hacksaw that perform the cutting operation is called

- A. handle
- B. Saw Blade +++
- C. Pin Blade

D. wing nut

80. Why we center punch before drilling a hole is

A. hammer/punching

B. to bore a hole

C. to avoid braking of the drill bit or wobbling +++

D. to drill a hole

81. is a tool used for marking lines parallel to the edge of wooden piece.

A. Mortise gauge

B. Marking gauge +++

C. Try square

D. Steel rule

82. Safety precautions are

A. constitution

B. Moral law

C. rules and regulations for students only

D. laws guiding practice in industry +++

83. Which of the snips is used for general purpose

A. Straight

B. Universal +++

C. Curved

D. Concave

84. Rolling machine is used for the generating of the following surfaces EXCEPT

A. two surface at right angle +++

B. cylindrical

C. circular

D. three surface at right angle

85. the screw of a fitters bench vice is made up of which thread

A. Vee thread

B. square or buttress thread +++

C. Square vee thread

D. acme thread

86. The vice that is always mounted on the drilling machine, power hacksaw etc is called

- A. Table vice
- B. pin vice
- C. Machine vice +++
- D. leg vice

87. Hammers are use for the following purposes EXCEPT

- A. driving pins
- B. striking other tools
- C. driving pulley
- D. marking out operation +++

88. The soft hammer head is made from the following materials EXCEPT

- A. Forge steel +++
- B. Forge Wood
- C. Forge rubber
- D. synthetic rubber

89. Taps are made from which materials

- A. Iron Hardened Steel
- B. Copper Hardened Steel
- C. Hardened and tempered steel +++
- D. Cast iron stell

90. Tap comes in a

- A. Set of two
- B. Set of Three +++
- C. Set of four
- D. none of the above

91. Which type of thread does the NUT has?

- A. Internal +++
- B. External
- C. Double thread
- D. Bi-external

92. The following tools are for marking operation EXCEPT

- A. Scriber
- B. Engineers square
- C. Steel rule +++
- D. Vee-block

93. Cutting operation cannot be performed in the fitters shop with

- A. Surface Plate +++
- B. Snip
- C. Hacksaw
- D. File

94. Micrometers screw gauge can measure accurately than

- A. Engineer Square
- B. Vernier Caliper +++
- C. Surface Caliper
- D. None of the above

95. Snips are specified by theiroverall length

- A. Overall Weight
- B. Overall Height
- C. Overall Length +++
- D. All of the above

96. Sharpening of the edges of universal snip blade is done at angle.....

- A. 5° +++
- B. 35°
- C. 15°
- D. 25°

97. Pins/Pegs of hack saw is used for.....

- A. Tightening bolt
- B. Tightening wing nut
- C. Holding the wing nut in position
- D. Holding the blade in position. +++

98. Irregular holes in thin metal can be produced using.....

- A. Abrafiles +++

B. Coping Saws

C. Piercing Saws

D. Junior Saws.

99. Which of the following type of chisels are used for cutting in corners and for cutting small oil grooves?

A. Cross cut chisel

B. Flat chisel

C. Half round chisel

D. Diamond point chisel +++

100. All the following are parts of a typical hacksaw EXCEPT.....

A. Face

B. Wing Nut

C. Frame +++

D. Hack saw blade.

101. is described as the technological process of making wooden components

A. Joining

B. Construction +++

C. Carpentry

D. Furniture

102. Dies are used for making

A. Internal Thread +++

B. External Thread

C. Semi-External Thread

D. Semi-Internal Thread

103. Files can be classified according to the following EXCEPT.....

A. Size

B. Type of cut of teeth

C. Shape

D. Hardness +++

104. Hammer is made of.....

- A. Hardened steel
- B. Forged steel +++
- C. Hardened high carbon steel
- D. High speed steel.

105. Which of the following is not a typical part of a screw driver.

- A. Base
- B. Tang +++
- C. Shan
- D. Handle

106. The abrasive on the wheel of a grinding machine is made of

- A. None of the Alternatives
- B. Carbonrundum or Aloxite +++
- C. Hardened Carbon
- D. Stainless Steel

107. can only be used for cleaning both tools and work piece externally.

- A. Cleaning tools +++
- B. Grinding tools
- C. Surface tools
- D. All of the above

108. Which of these is not a cleaning tool?

- A. Wire brush
- B. Carbon brush
- C. Bristle brush
- D. Cable brush +++

109. do not give full enclosure of the hexagonal head

- A. Ring Spanners +++
- B. Flat Spanners
- C. Combination Spanners
- D. Excel Spanners

110. is used to correct errors in machined components caused by warping or distortion.

- A. Scrapers +++
- B. Files
- C. Chisel
- D. Punch

111. According to roughness file is used for general purpose shaping work.

- A. Super smooth
- B. Bastard +++
- C. Second cut
- D. Smooth

112. is a measuring tool used for reading both internal and external diameter of thin cylindrical object.

- A. Micrometer screw gauge
- B. Slip gauge
- C. Vernier caliper +++
- D. Rollers

113. The part of a hacksaw that controls the frame of the hacksaw is known as

- A. Wing Nut +++
- B. Peg
- C. Pinning
- D. None of the above

114. The hole on the surface of an anvil which is wider than the punch hole is called thehole.

- A. Clearance
- B. Handle +++
- C. Punch
- D. All of the above

115. pein hammer is a general-purpose hammer.

- A. Straight
- B. Curve
- C. Ball +++
- D. Cross

116. Steel headed is used in conjunction with chisel during chiseling or chipping operation
- A. Hammer +++
 - B. Steel Mallet
 - C. Filling Hammer
 - D. None of the above
117. is used for cutting internal thread on cylindrical surface.
- A. Tapper
 - B. Tap +++
 - C. Die
 - D. All of the above
118. saw is used for the cutting of soft materials object like wood.
- A. Coping +++
 - B. Piercing
 - C. Junior
 - D. Abrasive file
119. is used to grip firmly the material or job on which work is being done.
- A. Bench
 - B. Swivel
 - C. Vice +++
 - D. Holder
120. is the name given to the wood obtained from well grown trees.
- A. Fibre
 - B. Log
 - C. Wood
 - D. Timber +++
121. Which of the following is not a type of hacksaw?
- A. Standard
 - B. Bastard +++
 - C. Abrasive file

D. Piercing

122. Chisels are forged from

A. Forged carbon steel +++

B. Nickel-alloy steel

C. Carbon-Iron steel

D. High carbon steel +++

123. Which of the following is not a part of a hammer?

A. Head

B. Edge +++

C. Handle

D. Pein

124. Which of the following is not a type of file according to roughness?

A. Second cut

B. Double Cut +++

C. Double Smooth

D. Rough Cut

125. Which of the following is not a cleaning tool?

A. Wire +++

B. Scraper

C. Brush

D. Wheel

126. Anvil is composed of a steel body with a hardened head.

A. Mild +++

B. Forged

C. Carbon

D. High Speed

127. The removal of thick layer of metal is known as

A. Chipping +++

B. Shaping

C. Threading

D. Clogging

128. Parts of typical hacksaw include the following EXCEPT.....
- A. Hacksaw blade
 - B. Frame
 - C. Wing nut
 - D. Tong +++
129. Which of the following is not a typical part of a Screw driver.....
- A. Shan
 - B. Base
 - C. Face +++
 - D. Handle
130. Dies are in thread cutting for cutting
- A. Internal and External thread
 - B. Internal thread
 - C. External thread +++
 - D. None of the above

SECTION B: AUTOMOBILE

131. The main two types of internal combustion engine are?
- A. Spark ignition and compression ignition +++
 - B. Spark ignition and compressor ignition
 - C. Press ignition and compression ignition
 - D. Start ignition and compression ignition
132. The following are the maintenance activities carried out regularly on an automobile except
- A. Checking of oil level in the sump
 - B. Checking the water/coolant level
 - C. Removal of spark plugs +++
 - D. Washing of the car
133. ----- switches the current to the spark plug through the ignition coil on and off.
- A. Contact-breaker +++
 - B. Battery
 - C. Alternator
 - D. Generator
134. ----- is necessary to ensure that vehicles operation remains unaltered or is restored to its original state.
- A. Cleaning
 - B. De-carbonization
 - C. Lubrication
 - D. Maintenance +++
135. ----- is used to measure the specific gravity of battery electrolyte.
- A. Hygrometer
 - B. Pyrometer
 - C. Hydrometer +++
 - D. Thermometer
136. is used as a work holding device in a carpenters workshop.
- A. Carpenters vice +++

- B. Carpenters clamp
- C. Carpenters gauge
- D. Plane gauge

137. ----- produces power in a car.

- A. Vehicle
- B. Engine +++
- C. Motor
- D. Machine

138. The carrier of electrons in the battery is called-----

- A. Water
- B. Acid
- C. Fuel
- D. Electrolyte +++

139. ----- is the removal of carbon deposit?

- A. Hydrogenation +++
- B. Chlorination
- C. De-carbonization
- D. Halogenation

140. Replacement of worn out part in the engine is called-----

- A. Maintenance
- B. Service
- C. Overhauling +++
- D. Repair

141. ----- uses spark plug for the ignition of explosive mixture of fuel and air?

- A. Diesel engine
- B. Petrol engine +++
- C. Kerosene engine
- D. Power engine

142. ----- is a self-propelled vehicle for the transportation of passengers and goods

- A. Aeroplane

- B. Ship
- C. Lorry
- D. Automobile +++

143. ----- is a container on wheels that carries all necessary components for operation.

- A. Drive shafts
- B. Fuel engine
- C. Vehicle +++
- D. Wheels

144. Another name for a diesel engine is-----

- A. Spark ignition engine
- B. Compressor ignition engine
- C. Fuel ignition engine
- D. Compression ignition engine +++

145. The electrolyte of a charged battery has the specific gravity value of-----

- A. 1.408
- B. 1.480
- C. 1.840 +++
- D. 1.804

146. is a tool that uses hard blade or wire with an abrasive edge to cut through softer materials.

- A. Plane
- B. Saw +++
- C. Clamp
- D. Vice

147. ----- holds the brake pad in the vehicle's braking system.

- A. Brake Caliper +++
- B. Brake disc
- C. Brake Fluid
- D. Clutch disk

148. Charging is the process of restoring a battery to its original state of

- A. Specific gravity +++
- B. Specific power
- C. Specific graphite
- D. Specific energy

149. ----- is used to charge the battery in the vehicle

- A. Generator
- B. Commutator
- C. Alternator +++
- D. Selector

150. The solution of Concentrated H_2SO_4 and Distilled water is -----

- A. Electrode
- B. Electrolyte +++
- C. Electron
- D. Electrolysis

151. ----- convert kinetic energy possessed by the vehicle into heat energy by means of friction

- A. Tyre
- B. Gear
- C. Clutch
- D. Brake +++

152. The valve that opens during induction stroke is called-----

- A. Inlet valve +++
- B. Exhaust Valve
- C. Opening Valve
- D. Outlet Valve

153. The volume of the mixture of fuel and air is decreased to ----- original volume during compression stroke.

- A. 2/7 times
- B. 3/7 times
- C. 1/7 times +++
- D. 4/7 times

154. ----- transform the low voltage into the high voltage needed to discharge across the spark plug gap as a spark.
- A. Ignition coil +++
 - B. Alternator coil
 - C. Distributor coil
 - D. Battery
155. The battery supply low in the vehicle
- A. Power
 - B. Voltage +++
 - C. Current
 - D. Resistance
156. ----- is used to store water in the vehicle for cooling the engine
- A. Water pump
 - B. Radiator +++
 - C. Regulator
 - D. Water reservoir
157. ----- links the piston to the crankshaft
- A. Connecting Shaft
 - B. Connecting Bolt
 - C. Connecting Pin
 - D. Connecting rod +++
158. ----- produces initial impetus to set the engine in motion
- A. Starter motor +++
 - B. Flywheel
 - C. Camshaft
 - D. Key
159. ----- compresses the mixture of fuel and air in petrol engine.
- A. Cylinder
 - B. Clutch
 - C. Piston +++
 - D. Connecting rod

160. ----- ignites the explosive mixture of fuel and air inside the cylinders to produce power

- A. Spark Plug +++
- B. Insulator
- C. Electrode
- D. Terminal

161. ----- is used to set the spark gap of the spark plug

- A. Feeler gauge +++
- B. Peeler gauge
- C. Plug Spanner
- D. Chisel

162. ---- of the spark plug determines how far the electrode protrudes into the combustion chamber

- A. Insulator
- B. Reach +++
- C. Gasket
- D. Wrench flat

163. The higher the insulator, the hotter the plug's operating-----

- A. Energy
- B. Temperature +++
- C. Heat
- D. Power

164. Which of the following feature is not part of the ignition system?

- A. Diode +++
- B. Battery
- C. Contact breaker
- D. Distributor

165. Engine block is defined as

- A. A metal casting that serves as a basic chassis on which vehicle is built
- B. A metal casting that serves as a basic structure on which other engine parts are installed

+++

- C. A metal casting that houses other components for better performance
- D. A metal casting that houses the main vehicular structure and components

166. converts petrol in fine spray and mixes air in proper ratio.

- A. Air-flow meter
- B. Aerometer
- C. Carburetor +++
- D. Distributor

167. regulates the opening and closing of valve in the piston engines

- A. Crankshaft
- B. Camshaft +++
- C. Connecting rod
- D. Gudgeon pin

168. converts the up and down movement of the piston into rotating motion.

- A. Crankshaft +++
- B. Camshaft
- C. Connecting rod
- D. Gudgeon pin

169. is used for joining rotating wheel to a reciprocating shaft.

- A. Camshaft
- B. Connecting rod +++
- C. Crankshaft
- D. Gudgeon pin

170. Engine block can be made from all of the following engineering materials except

- A. Al alloys
- B. Gray cast iron
- C. Ferrous alloys
- D. Forged steel +++

171. Internal Combustion Engine can be classified by the followings except
- A. Method engine design
 - B. Working cycle
 - C. Method of wheel positioning +++
 - D. Method of ignition
172. Which of the following operational processes is characterized by the movement of the piston from the Top Dead Centre (TDC) to Bottom Dead Centre (BDC) thereby making the inlet valve to be opened and the exhaust valve to be closed?
- A. Compression stroke
 - B. Power stroke
 - C. Induction stroke +++
 - D. Exhaust stroke
173. Which of the following operating cycle of the engine gives the definite sequence process combination?
- A. Induction, power, compression and exhaust strokes
 - B. Induction, power, exhaust and compression strokes
 - C. power, compression, Induction and exhausts strokes
 - D. Induction, compression, power and exhaust strokes +++
174. 100cc of air/fuel mixture is released into the cylinder through the inlet valve, what is the volume of the air/fuel mixture left for ignition.
- A. 30cc +++
 - B. 3.0cc
 - C. 13.0cc
 - D. 170cc
175. A typical modern tyre has a marking of “**185/70SR15**”. What is the diameter of the rim of the tyre?
- A. 18
 - B. 70
 - C. 13
 - D. 15 +++

176. A vehicle has a VIN of **4T1BE32K43U777252** on its chassis. What is the year of model of this vehicle?
- A. 2004 +++
 - B. 2003
 - C. 2007
 - D. 2005
177. Any machine which derives heat energy from the combustion of fuel and convert part of this energy into mechanical work is known as
- A. Petrol engine
 - B. Turbo engine
 - C. Heat engine +++
 - D. Gas engine
178. is used for starting and lightening in the vehicle.
- A. Battery
 - B. Carburetor
 - C. Spark plug +++
 - D. Ignition coil
179. The more the weight or load in a vehicle, the more the fuel consumed.
- A. True +++
 - B. False
 - C. All of the above
 - D. None of the above
180. The diagram shown in Figure below is called



- A. Camshaft
 - B. Crankshaft +++
 - C. Connecting Shaft
 - D. Piston
181. The diagram shown in Figure below is called



- A. Camshaft
- B. Connecting rod +++
- C. Crankshaft
- D. Piston

182. The diagram shown in Figure below is called



- A. Camshaft
- B. Crankshaft
- C. Connecting rod
- D. Camshaft +++

183. The component below is called.....



- A. Camshaft
- B. Crankshaft
- C. Connecting rod +++
- D. Connecting shaft

184. If the engine coolant leaks into the engine oil, it makes the engine oil

- A. Watery
- B. Foamy +++
- C. Black
- D. Milky

185. The motion of opens and closes valves in an engine.

- A. Crankshaft
- B. camshaft +++
- C. Timer
- D. Gudgeon pin

186. The purpose of a cylinder head gasket is to

- A. prevents the combustion gases from leaking from the joint between the cylinder block and cylinder head. +++
- B. prevent engine oil from going into the combustion chamber.
- C. prevent cooling water from flowing into the combustion chamber.
- D. prevent engine “knock” phenomenon.

187. The stroke of an engine is the

- A. distance between T.D.C. and B.D.C +++
- B. depth of the engine cylinder
- C. volume of the engine cylinder
- D. distance between T.D.C. and S.D.C

188. When the spark plug deposit indicates black coating of soot, it shows that the engine has been operating on.....

- A. Mixture too rich in oxygen
- B. Stoichiometric mixture
- C. Mixture too rich in fuel +++
- D. Adiabatic mixture

189. When air pockets are trapped in the hydraulic brake fluid line, it causes brake failure. The process of removing this air pockets is called

- A. De-pocketing
- B. Gleaning
- C. Sucking
- D. Bleeding +++

190. Modern engine blocks are made from aluminum alloy as against cast iron because

- A. material cost is lower and it can be casted more easily
- B. for uniformity since pistons are also made of aluminum

- C. it is lighter and have good heat dissipation property +++
- D. it has better strength mechanically and does not allow leakage

191. The air gap between the central electrodes of a spark plug is usually about

- A. 0.05 mm
- B. 0.5 mm +++
- C. 0,05 cm
- D. 0.5 cm

192. The ease with which petrol vapourizes is called.

- A. Oxidation
- B. Volatility
- C. Octane number +++
- D. Butane number

193. The function of the master cylinder is to

- A. convert brake pedal force to hydraulic pressure +++
- B. convert throttle pedal force to hydraulic pressure
- C. convert accelerator pedal force to hydraulic pressure
- D. none of the above

194. ----- reduces voltage arc and improve high voltage of the ignition coil in the ignition system.

- A. Plug
- B. Spark
- C. Capacitor +++
- D. Distributor

195. ----- distributes high voltage to each spark plug in each cylinder in turns.

- A. Battery
- B. Distributor +++
- C. Plug
- D. Capacitor

196. Two valves that open and close are-----

- A. Inlet and Exhaust +++
- B. Onlet and Outlet

C. Inlet and Outlet

D. Uplet and Downlet

197. ----- stores power developed by the engine

A. Radialwheel

B. Downwheel

C. Flywheel +++

D. Drivewheel

198. Another name for a petrol engine is -----

A. Spark ignition engine +++

B. Plug ignition engine

C. compression ignition engine

D. fuel ignition engine

199. ----- is used to transfer water from the radiator to the engine for cooling

A. Fuel pump

B. Water hose

C. Water pump +++

D. Oil pump

200. ----- stores fuel in the vehicle

A. Fuel tank +++

B. Fluid sump

C. Water tank

D. Oil sump

201. ----- stores lubricating oil in the vehicle

A. Grease sump

B. Water sump

C. Oil sump +++

D. Oil pump

202. ----- transfers lubricating oil from the oil sump to the lubricating part.

A. Oil pump +++

B. Oil sump

C. Water pump

D. Grease pump

203. ----- regulates the temperature of the engine

A. Hygrometer

B. Thermometer

C. Thermistor

D. Thermostat +++

204. ----- prevents water in the radiator not to reach boiling point

A. Pressure cap +++

B. Radiator cap

C. Temperature cap

D. Radiator cover

205. ----- cools the hot water in the radiator

A. Radiating Fan

B. Fan blade

C. Cooling fan +++

D. Ceiling fan

206. Petrol engine uses ----- fuel system

A. Carburetion +++

B. Injection

C. Respiration

D. Circulation

207. ----- Filters dirt from the lubricating oil in the vehicle engine

A. Oil Filter +++

B. Grease Filter

C. Candle Filter

D. Water Filter

208. Diesel engine uses ----- fuel system

A. Carburetion

B. Injection +++

C. Respiration

D. Circulation

209. ----- removes dirt from the fuel in the vehicle engine
- A. Oil Filter
 - B. Grease Filter
 - C. Candle Filter
 - D. Fuel Filter +++
210. ----- screen dirt particles from the air in the vehicle engine
- A. Oil Filter
 - B. Air Filter +++
 - C. Grease Filter
 - D. Fuel Filter
 - E. Water Filter
211. The following are the parts of the spark plug expect
- A. insulator
 - B. gasket
 - C. terminal
 - D. Spark cap +++
212. ----- prevents overheating of the engine
- A. cylinder gasket +++
 - B. cylinder head
 - C. cylinder engine
 - D. cylinder jacket
213. One of the functions of the lubricating oil is to -----
- A. reduce friksion
 - B. reduce friction +++
 - C. reduce fusion
 - D. Reduce friction
214. ----- disconnects the engine from the transmission for the smooth engagement of the drive
- A. clutch +++
 - B. gear
 - C. plate

D. brake

215. ----- allows the engine to run at different relative speed to the road wheels

A. selector

B. gear +++

C. pedal

D. clutch

216. ----- is used to control the movement of the vehicle

A. steering +++

B. tyre

C. rim

D. wheel

217. Two types of tyre that we have are -----

A. hard and soft

B. radial and cross-ply +++

C. alloy and Steel

D. smooth and rough

218. A typical modern tyre marking is 18570SR13, 185 means the tyre

A. width in millimeters +++

B. percent of its width

C. radial type

D. diameter

219. A typical modern tyre marking is 18570SR13, 70 means

A. width in millimeters

B. percent of its width +++

C. radial type

D. diameter

220. A typical modern tyre marking is 18570SR13, R means that the tyre is

A. width in millimeters

B. percentage of its width

C. radial type +++

D. diameter

221. The green colour in the hydrometer to test for the specific gravity of battery electrolyte indicates that the battery is.....

- A. good +++
- B. bad
- C. fair
- D. very fair

222. The white colour in the hydrometer to test for the specific gravity of battery electrolyte indicates that the battery is.....

- A. good
- B. bad
- C. weak +++
- D. not ok

223. The red colour in the hydrometer to test for the specific gravity of battery electrolyte indicates that the battery is.....

- A. good
- B. bad +++
- C. fair
- D. very fair

224. What is the proper word among this pair of words? (Hand brake; Parking brake; service brake; stopping)

- A. Hand brake
- B. Parking brake +++
- C. Service brake
- D. Parking & Service brake

225. A typical modern tyre has a marking of “185/70SR15”. What is the width of the tyre?

- A. 70
- B. 185 +++
- C. 70S
- D. 15

226. is a device for delivering electric current from an ignition system to the combustion chamber of a spark-ignition engine.

- A. Battery +++
- B. alternator
- C. spark plug
- D. wire conductor

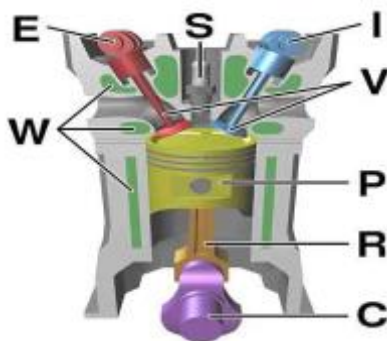
227. A cold spark plug is used in engines operated at high speeds.

- A. FALSE +++
- B. TRUE
- C. Not Sure
- D. None of the Above

228. Hot spark plug has..... insulator tip that prevents heat transfer into the water jackets

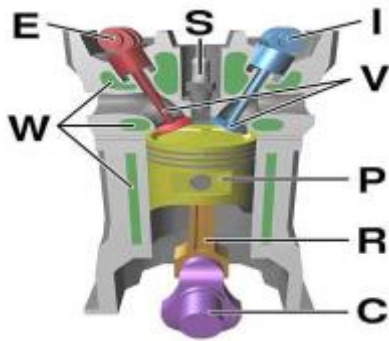
- A. a shorter
- B. a long +++
- C. Short and Long
- D. None of the above

229. What is the name of the component labeled S in Figure below.



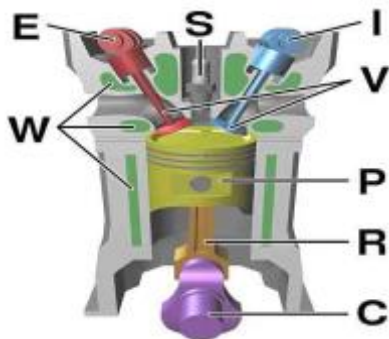
- A. Piston
- B. spark plug +++
- C. cylinder
- D. inlet valve

230. What is the name of the component t labeled V in Figure below.



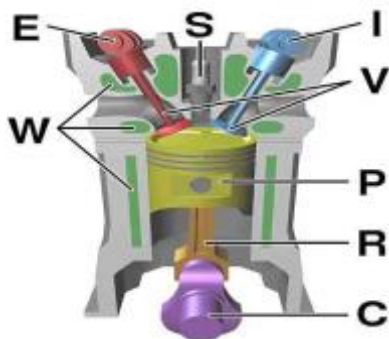
- A. Piston
- B. spark plug
- C. cylinder
- D. valves +++

231. What is the name of the component labeled **E** in Figure below.



- A. Piston
- B. spark plug
- C. cylinder
- D. exhaust valve +++

232. What is the name of the component labeled **I** in Figure below.



- A. Piston

- B. spark plug
- C. cylinder
- D. inlet valve +++

233. Four stroke spark ignition engines require ----- piston strokes to complete one cycle

- A. One
- B. two
- C. three
- D. four +++

234. The more speed of a vehicle the more the fuel consumption rate.

- A. FALSE
- B. TRUE +++
- C. Not Sure
- D. None of the Above

235. All options given below indicate the symptom for a battery in a low state of charge except

- A. Low hydrometer reading
- B. lack of power when starting
- C. burnt out bulbs +++
- D. poor light from the lamps

236. The Consists of a differential unit which provides final gear reduction to increase out put torque.

- A. Rear Axle +++
- B. Propeller Shaft
- C. Gear Box
- D. None of the above

237. is a hollow shaft that transmit power from the gear box to the rear axle.

- A. Rear Axle
- B. Propeller Shaft +++
- C. Gear Box

D. None of the above

238. A system that converts the heat energy in fuel to mechanical power is

A. Fuel Ignition

B. Fuel Engine

C. Heat Engine +++

D. None of the above

239. provides a suitable electrical spark to initiate combustion during the power stroke

A. Ignition System +++

B. Spark system

C. Clutch system

D. Brake System

240. The purpose of is to ensure that the spark is initiated at the precise time to the correct cylinder.

A. Valve timing

B. Chain timing

C. Ignition timing +++

D. Clutch timing

241. The type of tyre that has 2 to 3 layers of cords at an angle of between 80° and 90° is

A. Cross Ply

B. Radial Ply +++

C. Straight Ply

D. Random Ply

242. The charging system involves the following EXCEPT

A. Alternator

B. Wiring

C. Battery

D. Clutch +++

243. The lubrication system consists of the following EXCEPT

- A. Oil Separator +++
- B. Oil Filter
- C. Oil Sump
- D. Oil Pump

244. The following are types of saw except

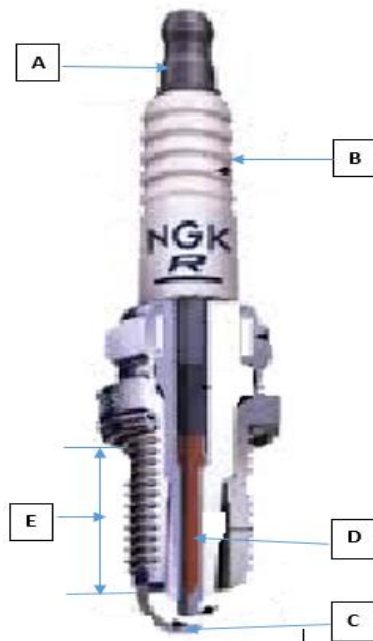
- A. Coping saw
- B. Mopping saw
- C. Keyhole saw
- D. Dovetail saw

245. The piston inside the cylinder is connected to the crankshaft via

- A. Connecting rod +++
- B. Camshaft
- C. Spark plug
- D. Flywheel

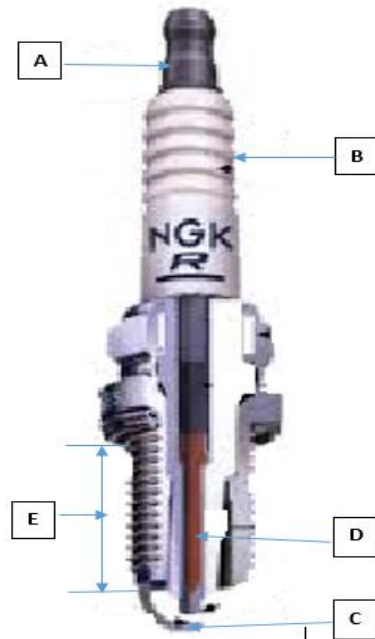
246. The part labelled A in the diagram below is called

- A. Terminal nut +++
- B. Terminal thread
- C. Corrugation
- D. Hexagon



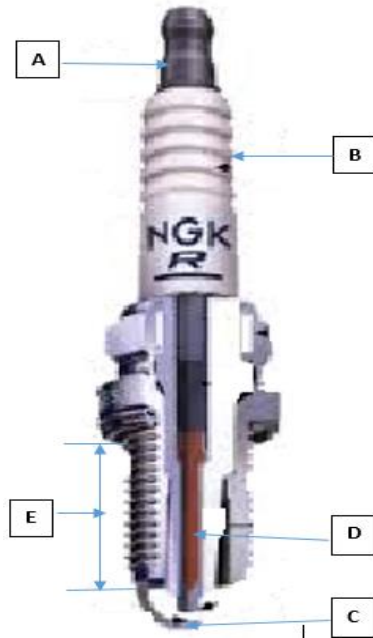
247. The part labelled B in the diagram below is called

- A. Insulator +++
- B. Corrugation
- C. Terminal
- D. Resistor



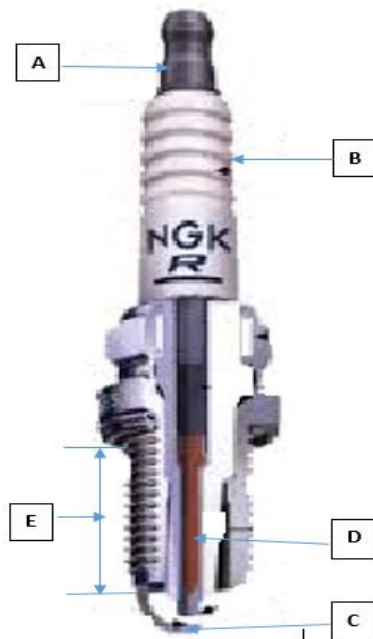
248. The part labelled C in the diagram below is called

- A. Metal Shell
- B. Centre Electrode
- C. Ground Electrode +++
- D. Spark Cap



249. The part labelled D in the diagram below is called

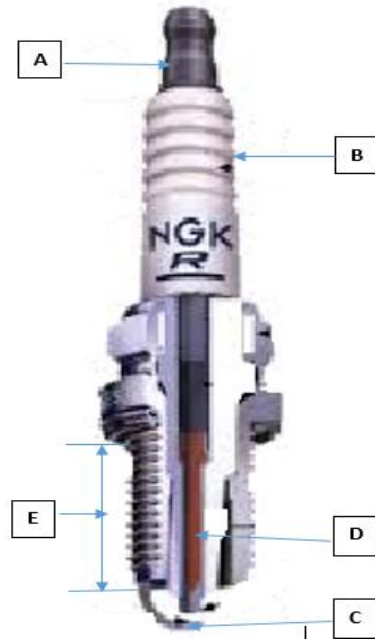
- A. Ground electrode
- B. Centre electrode
- C. Copper core +++
- D. Metal core



250. The part labelled E in the diagram below is called

- A. Metal cap

- B. Reach +++
- C. Centre electrode
- D. Spark gap



251. An automobile is slowed down or brought to rest by the application of
- A. Clutch
 - B. Turtle
 - C. Brake +++
 - D. Gear
252. The operating cycle of an engine consist of the following strokes
- A. Power , compression, exhaust, outlet
 - B. Induction, power, inlet, compression
 - C. Induction, compression, power, exhaust+++
 - D. Induction, spark, power, exhaust
253. Decarbonisation is the removal of carbon deposits in the part of an engine.
- A. Top cylinder head +++
 - B. Engine block
 - C. Carburetor
 - D. Top battery head
254. The main two types of internal combustion engine are

- A. Spark ignition and compression ignition +++
- B. Press ignition and compression ignition
- C. Start ignition and compression ignition
- D. Spark ignition and power ignition

255. The function of a battery in an automobile include the following except

- A. Supply of low voltage to the ignition coil
- B. Lighting of the headlamps
- C. Lightning of the interior lights in the automobile
- D. Opening of the engine +++

256. The following are the maintenance activities carried out regularly on an automobile except

- A. Checking of the water/coolant level
- B. Checking of oil level in the sump
- C. Removal of spark plugs +++
- D. Washing of the car

257. The marking of a tyre is given as “225/70SR16”. The width of the tyre in millimeters is

- A. 70
- B. 225 +++
- C. S
- D. 16

258. The rim diameter, in inches, suitable for a tyre with marking “225/70SR16” is

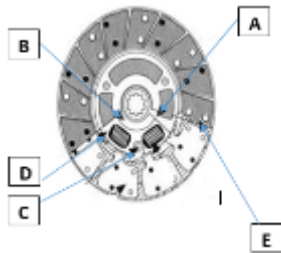
- A. 16 +++
- B. 12
- C. 13
- D. 225

259. Charging is the process of restoring a battery to its original state of

- A. Specific voltage
- B. Specific power
- C. Specific gravity +++
- D. Specific energy

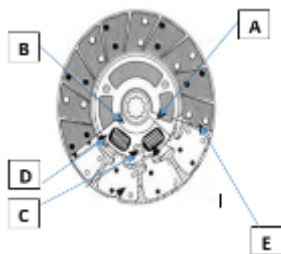
260. The part labelled A in the diagram below is called

- A. Drive washer +++
- B. Facing
- C. Hub flange
- D. Friction ring



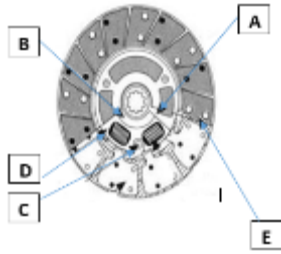
261. The part labelled B in the diagram below is called

- A. Drive washer
- B. Facing
- C. Hub flange
- D. Friction ring +++



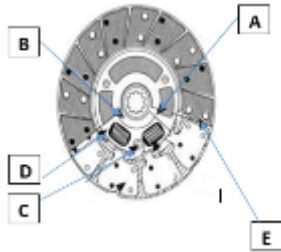
262. The part labelled C in the diagram below is called

- A. Drive washer
- B. Torsional coil spring +++
- C. Hub flange
- D. Friction ring



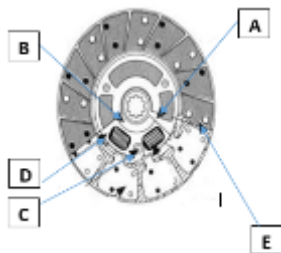
263. The part labelled D in the diagram below is called

- A. Torsional coil spring
- B. Hub flange +++
- C. Facing
- D. Drive washer



264. The part labelled E in the diagram below is called

- A. Facing +++
- B. Drive washer
- C. Torsional coil springs
- D. Hub flange



265. is a self-propelled vehicle for the transportation of passenger goods.

- A. Automobiles +++
- B. Combustion Engines

C. Internal Engines

D. Heat Engines

266. All the following are essential features of battery-coil ignition systems except

A. Distributor

B. Capacitor

C. Spark Plug

D. Camshaft +++

267. is the removal of carbon deposits in the top cylinder part of the engine.

A. De-ionization

B. Decarbonization +++

C. Overhauling

D. Exhaustion

268. The following are miscellaneous tools for wood work except

A. Mallet

B. Pincer

C. Claw hammer

D. Auger bit +++

269. A moving vehicle is slow down or brought to rest by the application of brakes by the conversion of mechanical energy possessed by the vehicle to

A. Heat energy +++

B. Chemical energy

C. Radial energy

D. None of the above

270. The following are symptoms of over-charged battery except

A. High hydrometer reading

B. Very frequent topping of battery

C. Poor light from the lamps +++

D. Burnt-out bulbs

SECTION C: REFRIGERATION & AIRCONDITIONING

271. Which of the following is an example of halogenated refrigerant?
- A. C_3H_8
 - B. NH_3
 - C. CCl_2F_2 +++
 - D. H_2O
272. Heat exchanger that removes heat from the hot refrigerant vapour and changes its phase back to liquid is?
- A. Evaporator
 - B. Expander
 - C. Refrigerant
 - D. Condenser +++
273. A system which can be used to achieve lower temperature than the surroundings is called
- A. Refrigerator
 - B. Air-conditioning
 - C. Refrigeration +++
 - D. Compressor
274. Refrigerating effect in the refrigerating system can be obtained from device called ...
- A. Evaporator +++
 - B. Evaporative cooling
 - C. Evaporative Heating
 - D. Evaporating Medium
275. The component in the Refrigeration system that is responsible for raising both the pressure and temperature of refrigerant vapour is called
- A. Compression
 - B. Expansion
 - C. Compressor +++
 - D. Evaporator

276. An instrument mostly used in refrigeration and air-conditioning is

- A. Vernier Caliper
- B. Dividing head
- C. Try square
- D. Manifold gauge +++

277. Refrigeration is achieved through the following methods except

- A. Vapour Jet
- B. Vapour Compression
- C. Vapour Adsorbent +++
- D. Vortex/Pulse system.

278. is regarded as the best method of detecting leakage in all refrigerating system.

- A. Electronic leakage detector +++
- B. Soap bubble method
- C. Halide torch method
- D. Sulphur candle method.

279. is a Heat transfer surface in which a volatile liquid is vaporized for the purpose of removing heat from the refrigerated space or product.

- A. Evaporative cooler
- B. Condensate
- C. Evaporator +++
- D. Shell and Tube

280. Which of the following is NOT part of compressor?

- A. Suction port.
- B. Discharge port
- C. Charging port
- D. Release port +++

281. is a process of cooling a metal very quickly.

- A. Quenching +++
- B. Normalizing
- C. Aging
- D. Annealing

282. The type of material used in the construction of the condensing unit is

- A. Iron +++
- B. Aluminium
- C. Copper
- D. Sif-bronze alloy

283. Which of the following does NOT apply to Refrigeration and Air-conditioning operation?

- A. Charging
- B. Flaring
- C. Brazing
- D. Facing +++

284. The following refrigerants are considered toxic to the environment except.....

- A. R-134a +++
- B. R-134
- C. R-12
- D. Ammonia

285. The system of refrigeration that source its power from natural power source is called ...

- A. Vapour Compression System
- B. Solar Refrigerator +++
- C. Vapour Jet
- D. Air Refrigeration

286. All these components are major components of the vapour compression system except ...

- A. Liquid receiver +++
- B. Condenser

- C. Expansion device
- D. Evaporator

287. A process that involves the use of energy and it is not a spontaneous process is called ...

- A. Cooling
- B. Air-conditioning
- C. Refrigeration +++
- D. Heat Pump

288. The treatment of air to control its temperature, humidity, cleanliness, odour and circulation is referred to as

- A. Air-conditioning +++
- B. Conditioned space
- C. Refrigerated Space
- D. An Enclosure

289. is the type of refrigeration and air-conditioning that involves vehicular, marine, vessels and air cargo transportation.

- A. Mobile Application +++
- B. Industrial Application
- C. Household Application
- D. Domestic Application

290. are suitable for cutting steel up to 1,25mm in thickness.

- A. Snips
- B. Guillotines +++
- C. Hacksaw
- D. Chiseling

291. One of these refrigerants is not a synthetic refrigerant.

- A. HFC
- B. CFC
- C. HCFC
- D. NH_3 +++

292. Refrigerants that consists of two or more different chemical compounds are called

- A. Mixtures +++
- B. Natural
- C. Synthetic
- D. Halogenated

293. The chemical formula $C_2H_2F_4$ represents a refrigerant called

- A. R-134
- B. R-600
- C. R-134a +++
- D. R-143a

294. The chlorofluorocarbon refrigerants are being replaced or phased out as they are found to

- A. Be Expensive
- B. Cause Ozone Layer Depletion +++
- C. Reacts with Products
- D. Consume More Energy

295. The following are considered as positive displacement type of compressor except

- A. Radial flow +++
- B. Rotary
- C. Reciprocating
- D. Acoustic

296. is a type of condensing unit that uses both air and water as its cooling medium.

- A. Air-cooled condenser
- B. Air-water cooled condenser
- C. Water-vapour condenser
- D. Evaporative condenser +++

297. is a device that reduces the condenser pressure to evaporator pressure and regulates the flow of refrigerant.

- A. Evaporator
- B. Expansion device +++
- C. Condenser
- D. Liquid receiver

298. Compared to hermetic compressor, an open compressor offers the following except

- A. Better compressor cooling
- B. Lower noise +++
- C. Serviceability and flexibility
- D. Higher efficiency

299. Air conditioning involves the following process except.....

- A. Heating
- B. Contraction+++
- C. Humidification
- D. Expansion

300. The main applications of refrigeration and air-conditioning in industries involve the following except.....

- A. Desalination of water
- B. Separation of gases
- C. Food processing
- D. Condensation of water +++

301. Condensers are classified as follows except.....

- A. Water Cooled condenser
- B. Natural and forced Condenser +++
- C. Air cooled Condenser
- D. Evaporative condenser

302. Which of the following is not a type of evaporator?

- A. Shell and coil type
- B. Flooded type
- C. Natural and forced Convection type +++
- D. Baudelot type.

303. The following are functions of evaporator in refrigerating systems except.....

- A. It aid refrigerating effect system
- B. Absorption of heat from the refrigerated space
- C. Rejection of heat from the refrigerated space +++
- D. Boils the refrigerant in the refrigeration system

304. Which of the following is not positive displacement type compressor?

- A. Centrifugal compressor
- B. Rotary vane compressor
- C. Direct and flat plate compressor +++
- D. Orbital compressor

305. The mass flow rate of refrigerant in the refrigeration is controlled by.....

- A. Condenser
- B. Evaporator
- C. Expansion device +++
- D. Compressor

306. Car radiator is of which of the following condenser type.....

- A. Air cooled
- B. Evaporative
- C. Water cooled
- D. Oil separator +++

307. Ozone layer depletion is mostly caused by which of the following type of refrigerants?

- A. Natural Refrigerant
- B. Hydrocarbons
- C. CFCs +++
- D. Organic (HCs)

308. The chemical formula of refrigerant R134a is?

- A. CHF
- B. CCl₃F

C. $\text{C}_2\text{H}_2\text{F}_4$ +++

D. CCl_2F_2

309. Refrigerant selection for a particular application in refrigeration and air conditioning system is based on the following properties except.

A. Environmental and safety properties

B. Economic properties

C. Ozone Depletion and Global Warming Potential properties +++

D. Thermo-Physical properties

310. Arrangement of compressor motor or external drive includes the following except.

A. Close type +++

B. Semi-hermetic

C. Open type

D. Hermetic type

311. The art of refrigeration can be achieved through the following type of system except..

A. Vortex Refrigeration system

B. Air- Refrigeration system

C. Vapour Jet Refrigeration system

D. Evaporation Refrigeration system +++

312. The diagram below is called

A. Liquid receiver

B. Dryer +++

C. Copper tube

D. Shell and coil



313. Condensation of vapour into liquid in condenser unit takes place by.....

A. Natural convection to the heat sink

B. Liquid expansion to the heat sink

C. Heat rejection to the heat sink +++

D. Heat absorption to the heat sink

314. The chemical formula of refrigerant R12 is.

A. CHF

B. CCl₃F

C. C₂H₂F₄

D. CCl₂F₂ +++

315. Heat exchanger that removes heat from the hot refrigerant vapour and changes its phase back to liquid is?

A. Evaporator

B. Expander

C. Refrigerant

D. Condenser +++

316. A system which can be used to achieve lower temperature than the surroundings is called

A. Thermo-electric

B. Pulse- jet system

C. Refrigeration system +++

D. Compression system

317. The component of refrigerating system responsible for moisture removal is called

A. Evaporator

B. Expander

C. Dryer +++

D. Condenser

318. In which of the following state can refrigerant exist in refrigerating system

A. Liquid & Vapour +++

B. Solid & Gas

C. Solid and Liquid

D. Gas & Vapour

319. The classification of compressor based on trapping a certain volume of gas in a cylinder that is continuously reduced by the compressing device is called

A. Positive displacement +++

- B. Roto-dynamic
- C. Hermetic sealed
- D. Working principles

320. Which of the following type of Air conditioning system finds application in refrigeration system?

- A. Winter Air conditioning
- B. Refrigeration
- C. Summer Air conditioning
- D. Cooling system +++

321. The type of compressor that houses both the motor and compressing housing inside the same steel shell welded together is called

- A. Positive displacement compressor
- B. Rotor-dynamic
- C. Hermetic compressor +++
- D. Open compressor

322. The correct way of designating CCl_3F refrigerant is?

- A. R12
- B. R13
- C. R22
- D. R11 +++

323. The Capillary tube

- A. reduces the high pressure in the liquid line to the low pressure in the evaporator +++
- B. equalizes the low pressure in the dryer liquid line with that of the discharge line in the condenser
- C. C. increases the high pressure in the liquid line to the high pressure in the evaporator
- D. equalizes the low pressure in the gaseous line to the low pressure in the condenser.

324. Compressor is a device for

- A. equalizing pressure between condenser and dryer
- B. increasing the pressure and temperature of the refrigerant +++
- C. creating pressure difference between condenser and evaporator
- D. causing phase change of refrigerant in Refrigeration and Air-conditioning systems

325. The flowing of refrigerant through a refrigerating system occurs in the following order:

- A. compressor-dryer-expansion valve-condenser-evaporator
- B. Compressor-condenser-expansion valve-dryer-evaporator
- C. compressor-condenser-expansion valve-evaporator-dryer
- D. compressor-condenser-dryer-expansion valve-evaporator +++

326. The likelihood of an undesired event with specified consequence occurring within a specified period, or in a specified environment is called

- A. Hazard
- B. Safety
- C. Accident +++
- D. Risk

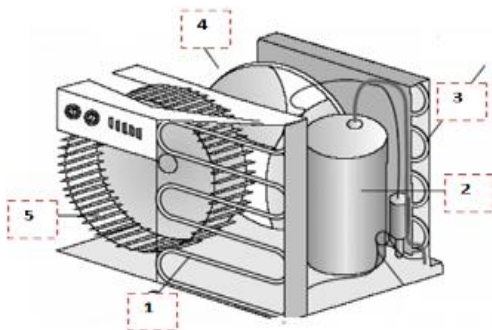
327. Charging of refrigerating/air-conditioning system is achieved with the use of

- A. Manifold gauge
- B. multifold gauge
- C. Manifold gauge +++
- D. Mamifold gauge

328. The appropriate method of joining tube in Refrigeration Air-conditioning system is

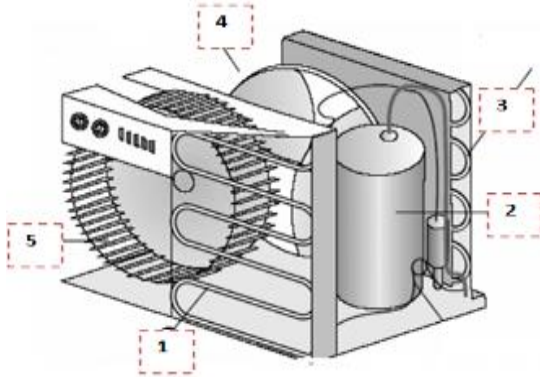
- A. Brazing +++
- B. Welding
- C. soldering
- D. smelting

329. The part labeled 5 in the figure below is called



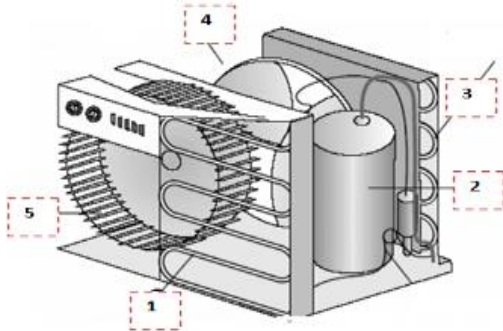
- A. Evaporator
- B. Condenser fan
- C. Evaporator fan +++
- D. Compressor

330. The part labeled 1 in the figure below is called



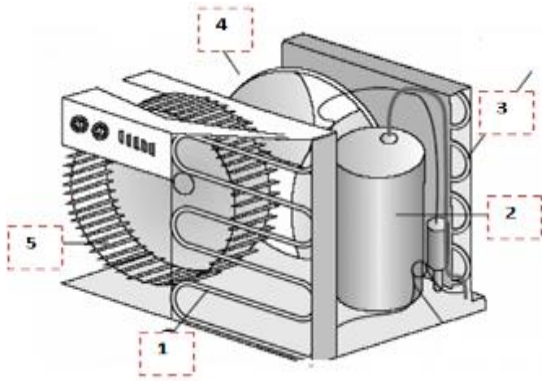
- A. Evaporator +++
- B. Condenser fan
- C. Evaporator fan
- D. Compressor

331. The part labeled 3 in the figure below is called



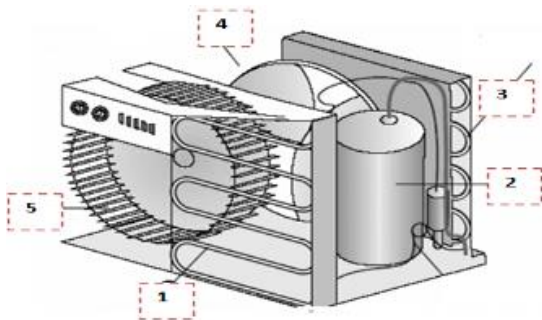
- A. Evaporator
- B. Condenser fan
- C. Evaporator fan
- D. Condenser +++

332. The part labeled 2 in the figure below is called



- A. Evaporator
- B. Condenser fan
- C. Evaporator fan
- D. Compressor +++

333. The part labeled 4 in the figure above is called



- A. Evaporator
- B. Condenser fan +++
- C. Evaporator fan
- D. Compressor

334. The type of flux used for brazing in refrigeration and air-conditioning operation is

- A. Sifgold
- B. Sifbronze +++
- C. Sifalloy
- D. All of the above

335. The figure below is used for



- A. Flushing
- B. Charging +++
- C. Brazing
- D. None of the above

336. The instrument in the figure above is called



- A. Sight Glass
- B. Manifold Gauge +++
- C. Pressure Gauge
- D. Compound Gauge

337. is the type of alloy used when joining iron to copper

- A. Brass rod +++
- B. Silverous alloy
- C. Aluminium alloy
- D. All of the above

338. The type of refrigeration application that is used for preservation is called

- A. Industrial
- B. House hold +++
- C. Mobile
- D. Commercial

339. One of the following components cannot be found in a typical refrigerator

- A. Blower +++
- B. Evaporator
- C. Condenser
- D. Throttle valve

340. The figure below is called



- A. Hermetic compressor +++
- B. Open Compressor
- C. Acoustic Compressor
- D. Rotary Compressor

341. The instrument in the figure below is called



- A. Multimeter +++
- B. Multi tester
- C. Voltage meter
- D. Ohm meter.

342. A system that employ both air and water as cooling media to extract heat from the refrigerant.

- A. Flooded condenser
- B. Shell and Coil condenser
- C. Evaporative condenser

D. Finned type condenser

343. The air-conditioning system controls the following parameters EXCEPT

A. Purity

B. Colour +++

C. Movement

D. Temperature

344. The name of the control in the figure below is called



A. Automatic Expansion Valve

B. Capillary Tube

C. Thermostatic Expansion Valve +++

D. Regulating valve

345. is considered as the most sensitive method of detecting leakages

A. Soap Bubble

B. Halide Torch

C. Sulphur Candle

D. Electronic detector +++

346. The set of kit used for soldering and brazing of pipes in refrigeration and air-conditioning practice is called

A. Oxygen/Acetylene Equipment +++

B. Oxygen/Carbon Equipment

C. Acetylene/Nitrogen Equipment

D. Flaring Equipment

347. The following are examples of thermometer EXCEPT

- A. Mercury-in-glass
- B. Minimum and Maximum
- C. Temperature recorder
- D. Upper and Lower thermometer +++

348. The type of refrigerant leakage detector which the flame changes from yellow to green when in contact with leaked refrigerant is

- A. Soap Bubble
- B. Halide Torch
- C. Sulphur Candle +++
- D. Electronic Detector

349. The TWO basic method of joining component parts of refrigeration and air-conditioning system together is called

- A. Flare connection/Brazing +++
- B. Welding/Brazing
- C. Flaring/Welding
- D. All of the above

350. The tool in the figure below is used for



- A. Cutting copper tubes +++
- B. Flaring Copper Tubes
- C. Sealing copper tubes
- D. Burring copper tubes

351. The name of the tool shown in the figure above is



- A. Manifold Set
- B. Brazing Set
- C. Flaring Set +++
- D. None of the above

352. are those liquids which are used for transporting thermal energy from one location to other.

- A. Secondary refrigerant +++
- B. Primary refrigerant
- C. Organic refrigerant
- D. Azeotropic refrigerant

353. What type of compressor is shown in the figure below?



- A. Centrifugal
- B. Reciprocating
- C. Rotary +++
- D. Screw

354. The figure below is an example of Evaporator



- A. Shell & Tube
- B. Plate +++
- C. Flooded
- D. Shell & Coil.

355. The pinch-off tool is used for

- A. Closing the orifice of a tube +++

- B. Opening the orifice of a tube
- C. Flaring the tube
- D. All of the above.

356. The name of the component in the figure below is



- A. Forced Draft-Air cooled condenser
- B. Natural Air cooled condenser +++
- C. Evaporator
- D. Fin Evaporator

357. The type of alloy used for the joining of copper to copper is called

- A. Silverous Alloy +++
- B. Copper Alloy
- C. Brass Alloy
- D. Aluminium Alloy

358. The method of joining metals together in refrigeration and air-conditioning practice without the application of heat is called

- A. Brazing
- B. Welding
- C. Flaring +++
- D. Electric welding

359. is a heat transfer surface in which a volatile liquid is vapourized for the purpose of removing heat from the product or space.

- A. Condenser
- B. An Evaporator +++
- C. An Expansion Device
- D. A Compressor

360. The major challenges in refrigeration and air-conditioning practice is

- A. Leakage +++
- B. Expansion
- C. Burnt out
- D. Flooding

361. The correct way of designating NH_3 refrigerant is?

- A. R-12
- B. R-717 +++
- C. R-714
- D. R-11

362. The mixture of two refrigerants to produce R500 series which cannot be separated is called

- A. Azeotropic +++
- B. Zeotropic
- C. Inorganic Refrigerants
- D. Halogenated Refrigerants

363. The direct refrigerant used in vapour compression system which is in direct contact with the space or material being refrigerated can be classified as?

- A. Systematic Refrigerant
- B. Secondary Refrigerant
- C. Best Refrigerants
- D. Primary Refrigerant +++

364. Which of the following statement(s) is/are true about the property of a good refrigerant?

- i. Good refrigerants should have low latent heat of vaporization
 - ii. Low C.O.P in the working temperature range
 - iii. Very toxic and flammable
 - iv. It should be able to easily deplete the ozone layer and contribute to global warming
 - v. It should have low specific heat and considerably high thermal conductivity
- A. All of the above
 - B. i, ii, and iv

- C. iii and iv
- D. v only +++

365. Which of the following is an example of secondary refrigerants

- A. Brine +++
- B. R-12
- C. Di-chloro di-fluoro methane
- D. Propane

366. Air in the air conditioning system is dehumidified when its temperature is?

- A. Heated spontaneously
- B. Cooled below dew point temperature +++
- C. Heated at high pressure
- D. Expanded

367. The process of using fan to aid the heat being ejected from the air cooled condenser can be simply described as

- A. Natural air cooled condenser
- B. Water cooled condenser
- C. Forced air cooled condenser +++
- D. Evaporative cooling

368. The application of refrigeration system in automobiles can be grouped under

- A. Household application
- B. Industrial application
- C. Commercial application
- D. Mobile application +++

369. Evaporative cooling can be best described as?

- A. The process of reducing the temperature of a substance by evaporation of its molecules
+++
- B. Cooling the body of fluid by the evaporator of a refrigeration system
- C. Cooling by salt solution
- D. All of the above

370. The correct method of numerical designation for inorganic refrigerants is **R_x**, where **x** is?

- A. $700 + \text{molecular weight of the compound} +++$
- B. $717 - \text{molecular weight of the compound}$
- C. $700 + \text{atomic mass of the compound}$
- D. $700 + \text{number of proton in the compound}$

371. The art of refrigeration can be obtained through the following refrigerating systems except

- A. Vapour compression system
- B. Thermo – electric system
- C. Vapour absorption system
- D. Isothermal system +++

372. The full meaning of C.O.P in refrigeration and air conditioning is?
- A. Coefficient of Practical
 - B. Correlation of Performance
 - C. Coefficient of Performance +++
 - D. Constant of Practice
373. Which among the following refrigerants can be used as a replacement in the modern refrigerating system because of the concern over ozone depletion of CFCs
- A. CHClF_2 +++
 - B. CCl_2F_2
 - C. CCl_3F
 - D. CClF_3
374. The system of refrigeration based on absorption by evaporation of one of the two substances by other which are kept in two separate vessels and without compressor in the system is called.....
- A. Vapour compression system
 - B. Thermo – electric system
 - C. Vapour absorption system +++
 - D. Vapour jet refrigeration system
375. The tool used for the lighting of the Oxy-Acetylene equipment is called
- A. Spark lighter +++
 - B. Spark Fire
 - C. Gasoline Fire
 - D. Refrigerant Fire
376. Change of state takes place in a refrigeration system in the
- A. evaporator and control
 - B. drier and receiver tank
 - C. condenser and evaporator +++
 - D. condenser and compressor
377. Which of the following is not a component of a vapour compression refrigeration system?
- A. absorber +++

- B. condenser
- C. compressor
- D. evaporator

378. Overload protector in a refrigeration system is located in the

- A. control panel
- B. liquid receiver
- C. evaporator outlet
- D. compressor terminal +++

379. Joining of refrigeration copper pipe is done by.....

- A. brazing +++
- B. welding
- C. screwing
- D. riveting

380. The merit of installing the suction pipe downwards is to.....

- A. reduce operation cost
- B. increase pumping efficiency
- C. Ensure effective discharge pressure
- D. enhance oil return +++

381. Which of the following control is popular with domestic refrigeration unit

- A. automatic expansion valve
- B. thermostatic expansion valve
- C. capillary tube +++
- D. high side float

382. Which of the following is not used for servicing a window air conditioner.....

- A. screw driver
- B. pliers
- C. pipe wrench +++
- D. Allen key

383. The component that changes refrigerant from liquid to vapour is.....

- A. control
- B. evaporator +++

C. Condenser

D. Drier

384. The condensing medium used in evaporative condenser is

A. air only

B. water only

C. both air and water +++

D. refrigerant

385. A condenser is used in Pressure side of a refrigeration system

A. low

B. high +++

C. both high and low

D. intermediate

386. The refrigerant supplied to a compressor must be

A. superheated vapour refrigerant +++

B. dry saturated liquid refrigerant

C. a mixture of liquid and vapour refrigerant

D. none of the above

387. The pressure at the inlet of a refrigerant compressor is called.....

A. suction pressure +++

B. discharge pressure

C. Critical pressure

D. back pressure

388. A compressor is used to.....

A. raise the pressure of the refrigerant

B. raise the temperature of the refrigerant

C. circulate the refrigerant through the refrigerating system

D. all of the above +++

389. The highest temperature during the cycle in a vapour compression refrigeration system occurs after.....

A. compression +++

B. condensation

- C. expansion
- D. evaporation

390. The sub-cooling is a process of cooling the refrigerant in vapour compression refrigeration system

- A. before compression
- B. after compression
- C. before throttling +++
- D. after throttling

391. is the control of temperature, humidity, air cleanliness and its distribution to meet the comfort requirements of human beings and industrial requirements.

- A. Comfort cooling
- B. Air-conditioning +++
- C. Refrigeration
- D. All of the above

392. The chemical name of Refrigerant 12 is

- A. Dichloromonofluoromethane
- B. Monochlorodifluoromethane
- C. Dichlorodifluoromethne +++
- D. None of the above

393. The full meaning of VCRS is

- A. Vast cooling refrigeration systems
- B. Vapour condensing refrigeration systems
- C. Vapour compressor refrigeration systems
- D. Vapour compression refrigeration systems +++

394. The following are the classification of positive displacement type of compressors except

- A. Reciprocating
- B. Radial flow +++
- C. Orbital
- D. Rotary

395. is the type of compressor that has the motor and the compressor enclosed in the same housing to prevent leakage.
- A. Hermetic +++
 - B. Semi-hermetic
 - C. Open Hermetic
 - D. Twin screw
396. is the place at which the desired refrigerating effect of a refrigerator or air-conditioner is achieved.
- A. Evaporator +++
 - B. Condenser
 - C. Expansion device
 - D. Compressor
397. The type of compressor that offers flexibility and serviceability is called
- A. Open type +++
 - B. Semi-hermetic
 - C. Hermetic
 - D. Twin screw
398. is the part of refrigeration system that houses both the refrigerant and the lubricating oil.
- A. Expansion device
 - B. Compressor +++
 - C. Condenser
 - D. Evaporator
399. and are the two methods of charging refrigerant into a system.
- A. Water and Vapour
 - B. Gas and vapour
 - C. Vapour and air
 - D. Liquid and vapour +++
400. All the following are methods of achieving refrigeration except
- A. Vapour absorption
 - B. Vapour compression

C. Vapour adsorption +++

D. Vapour jet

401. The name of the lubricant used for the lubrication of refrigeration compressor is called

A. Soluble oil

B. Capella D oil +++

C. Mineral oil

D. Fixed oil

402. The following are examples of synthetic refrigerant except

A. HC's +++

B. CFC's

C. HFC's +++

D. HCFC's

403. Condensation of vapour into liquid in condenser unit takes place by.....

A. Natural convection to the heat sink

B. Liquid expansion to the heat sink

C. Heat rejection to the heat sink +++

D. Heat absorption to the heat sink

404. Fully saturated, Azeotropic and Zeotropic refrigerants include.....

A. R12, R503 and R407 +++

B. R11, R503 and R407

C. R717, RR503 and R407

D. R178, R603 and R 503.

405. An evaporator is used in Pressure side of a refrigeration system

A. High

B. Low +++

C. Both High and Low

D. Intermediate

406. The pressure at the outlet of a refrigerant compressor is called

A. Suction Pressure

B. Discharge Pressure +++

C. Critical Pressure

D. Back Pressure

407. An expansion device is used to

A. Raise the pressure of the refrigerant

B. Raise the temperature of the refrigerant

C. Circulate the refrigerant through the refrigerating system

D. None of the above +++

408. The classification of compressor based on trapping a certain volume of gas in a cylinder that is continuously reduced by the compressing device is called.....

A. Positive displacement +++

B. Roto-dynamic

C. Hermetic sealed

D. Working principles

409. The following are type of flames except

A. Carbonated flame +++

B. Oxidizing flame

C. Carburizing flame

D. Neutral flame

410. What is the usual composition of soft solder?

A. Lead and antimony

B. Lead and Tin +++

C. Tin and Zinc

D. Zinc and copper

411. What is the base of most of silver brazing flux?

A. Vaseline

B. Zinc chloride

C. Borax +++

D. Sodium chloride

412. What is the advantage of silver brazing over soft soldering?

A. Cheaper

B. Quicker

- C. Neater
- D. Stronger +++

413. What is the approximate temperature at which silver brazing alloys flow?

- A. 618 °C +++
- B. 593 °C
- C. 565 °C
- D. 537 °C

414. Why must the flux be removed after soldering and brazing?

- A. It is not
- B. To help one see the quality of the job
- C. To prevent corrosion +++
- D. To polish the joint

415. How many joints are soldered when three pieces of tubing are joined with a coupling and a swagged joint?

- A. 2 +++
- B. 3
- C. 4
- D. 5

416. How do pipe thread seal together?

- A. Cut on a taper
- B. Gasket +++
- C. "O" rings
- D. Each thread is tapered

417. How many wrenches should be used to install a coupling on a pipe?

- A. One
- B. Three
- C. Two +++
- D. One plier and one pipe wrench.

418. The melting point of ice is

- A. 273K +++
- B. 0°F

- C. 0° Rankin
 - D. 10°C
419. Why is the tubing cut with a cutter instead of being cut with a hacksaw?
- A. It is easier
 - B. To keep copper chips out of the refrigerant lines +++
 - C. It is faster
 - D. To keep air out of the tubing
420. What is the most frequent service operation performed on a condenser?
- A. Replacing it
 - B. Purging it
 - C. Straightening the fins
 - D. Cleaning it +++
421. What colour of the halide torch flame indicates a leak?
- A. Red
 - B. Blue
 - C. Yellow
 - D. Green
422. What instrument should be used to check for a short circuit in a system?
- A. Voltmeter
 - B. Ohmmeter +++
 - C. Ammeter
 - D. Wattmeter
423. How many terminals are used in most hermetic compressor?
- A. Three +++
 - B. Four
 - C. Two
 - D. Five
424. What is the recommended temperature range for a chest type food freezer?
- A. 0 °C - -15 °C +++
 - B. -28 °C - -23 °C
 - C. -34 °C - -31.6 °C

D. $-40^{\circ}\text{C} - 34.4^{\circ}\text{C}$

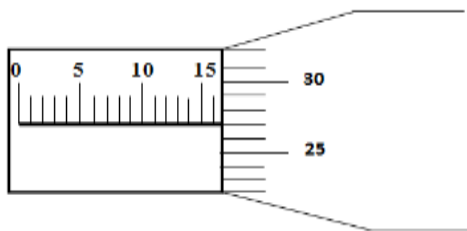
425. A heat pump is an example of
- A. Compressor
 - B. Refrigerator +++
 - C. Condenser
 - D. Water chiller
426. What type of refrigerant control is in common use on small domestic freezers?
- A. Low side float valve
 - B. Thermostatic expansion valve
 - C. Capillary tube +++
 - D. Automatic expansion valve
427. Why is a filter put at the inlet of the capillary tube?
- A. To remove oil
 - B. To remove air
 - C. To remove solid foreign particles
 - D. To remove moisture +++
428. What type of material is capillary tube made of?
- A. Copper +++
 - B. Aluminum
 - C. Stainless steel
 - D. Brass
429. Where is the most common place to install a drier in the system?
- A. Between the refrigerant control and the evaporator
 - B. In the liquid line +++
 - C. Between the compressor and condenser
 - D. Between the condenser and liquid receiver
430. The chemical formula of Ammonia refrigerant is
- A. R 717 +++
 - B. R 708
 - C. R 707

SECTION D: MACHINING

431. is a precision instrument which is used for measuring thickness and diameter of a work piece

- A. Vernier caliper
- B. micrometer screw gauge
- C. precision steel rule +++
- D. protractor head

432. The reading on the figure shown below is:



- A. 15.27 mm
- B. 15.37 mm
- C. 16.37 mm
- D. 16.27 mm +++

433. The ability of a material to resist indentation or deformation is

- A. ductility
- B. toughness +++
- C. hardness
- D. true fracture strength

434. Calculate the index arm setting required to give eighteen (18) equally spaced divisions. The index plate has the number of holes circle: 46, 47, 49, 51, 53, 54, 57, 58, 59, 62, 66.

- A. $3\frac{12}{58}$ +++
- B. $3\frac{12}{54}$
- C. $3\frac{6}{51}$

D. $3\frac{12}{62}$

435. is used to cool the cutting tool and the work piec

- A. Milky solution
- B. cutting fluid
- C. Oil +++
- D. grease

436. The Health and Safety at Work Act of 1974 is of special significance for those who work inbecause it places the responsibility for the maintenance of standards equally on the employer and employee.

- A. Construction/Company
- B. Workshop/Safety
- C. Engineering/Safety
- D. Machine Shop/Machine +++

437. The following operations can be perform on the lathe except

- A. Screw Cutting +++
- B. Taper Turning
- C. Knurling
- D. Gear Cutting

438. The following precautions should be observed in caring of milling cutters except....

- A. Support The Cutter Properly And Hold The Work Rigidly
- B. Use The Correct Cutting Speed And Feed
- C. Use The Correct Cutter For The Job +++
- D. Cutter's Direction Of Rotation in down-cut milling should be opposite to linear movement of the work piece

439. The principal constituents of the grinding wheel of a grinding machine are made of

- A. Stone and Cement
- B. Abrasive Grains and Bond
- C. Iron and Stone
- D. Carbide and Bond +++

440. All the following are reasons for grinding operation except to
- A. Sharpen tools
 - B. Machine very hard materials
 - C. Remove surplus materials +++
 - D. Shape materials
441. A tool that is used in order to attain good surface finish when shaping because it prevents chattering and digging in the tool into the work is
- A. Flat Nose Swan Necked Tool
 - B. Slot Cutting Tool
 - C. Straight Nose Tool
 - D. Cranked Tool +++
442. The controls the depth of cut and is also used for feeding when performing perpendicular, vertical, and angular operations on the
- A. Shaping Machine Tool Slide
 - B. Clapper Box +++
 - C. Tool Post
 - D. Cross Rail
443. Make 7 equal divisions slots on 60 mm diameter work pieces on the milling machine using the simple indexing head. The number of hole circles available on the index plate is: 6, 10, 12, 15, 18, 20, 21, 30, 36, 42, and 60. From the available hole circles given, 6, 12, 18, 30, 36, 42, and 60 are hole circles which can be used
- A. 12 holes
 - B. 42 holes +++
 - C. 32 holes
 - D. 52 holes
444. The two methods of cutting action of milling are andmilling.
- A. Up/Down-Cut +++
 - B. Down/Side-Cut
 - C. Up/Side-Cut
 - D. Vertical/Horizontal-Cut

445. operation is a process of producing conical symmetrical shapes on the lathe machine.

- A. Chamfering
- B. Spinning
- C. Knurling +++
- D. Taper Turning

446. Large diameter holes are produced on a lathe by:.....

- A. Step drilling
- B. Reaming +++
- C. Facing
- D. Boring

447. The work holding device on Lathe machine is mounted on the

- A. Carriage
- B. Spindle +++
- C. Compound slide
- D. Cross slide

448. Machine tools are classified based on the following except

- A. Cutting fluid
- B. Cutting agents
- C. Cutting speed
- D. Cutting feed +++

449. The abrasive on the wheel of a grinding machine is made of either

- A. Carbonrundum or aloxite +++
- B. Carbonrundum or harden steel
- C. Carbonrundun or plexolite
- D. None of the above.

450. Full meaning of PPE as regards safety rules is -----

- A. Personal People Equipment
- B. People Personal Equipment +++
- C. Personal Protective Equipment
- D. Protective Personal Equipment

451. The four- jaw chuck can also be called -----
- A. Three-jaw chuck
 - B. Semi-independent chuck
 - C. Universal scroll chuck
 - D. Independent chuck +++
452. Feed rate refers to:
- A. Spindle speed
 - B. Chuck rotation
 - C. Tool advancement on the work piece +++
 - D. Depth of cut on the work piece
453. ----- is used in drilling and countersinking centre holes in one operation
- A. Reaming nose tool
 - B. Round nose tool +++
 - C. Knurling nose tool
 - D. Centre drill tool
454. The most commonly used cutting fluid is called -----
- A. Soluble oil
 - B. Fixed oil +++
 - C. Mineral oil
 - D. Compound oil
455. ----- can be offset for longer external tapering
- A. The tailstock +++
 - B. Topside stock
 - C. Compound slide
 - D. Carriage stock
456. The part of the lathe machine that accommodate cutting tools is known as
- A. Tailstock
 - B. Tool post
 - C. Headstock +++
 - D. Independent chuck

457. is used for mounting work of awkward shapes which cannot readily be chucked.

- A. Faceplate
- B. Steady rest +++
- C. Universal scroll chuck
- D. Lead rest.

458. circulates filtered fluid usually directed in continuous flow over the workpiece at the tool point

- A. Drip-can
- B. Spraying pump
- C. Fluid gun
- D. Spraying gun +++

459. The requisite operation to be performed on a long work piece before carrying out parallel turning between centres on the lathe is the

- A. Chamfering
- B. Centre-drilling +++
- C. End-drilling
- D. Face-drilling

460. Grinding is done for the following purposes except

- A. To remove surface finish
- B. To obtain required surface finish +++
- C. To sharpen tools
- D. To taper surplus materials

461. The work piece on a Lathe machine rotates in direction

- A. Clockwise
- B. Semi clockwise +++
- C. Anticlockwise +++
- D. Perpendicular

462. is used to cool the cutting tool and the work piece

- A. Water
- B. Oil

- C. Cutting fluid +++
- D. Milky fluid

463. This shape cannot be produced with the lathe

- A. Conical
- B. Cylindrical +++
- C. Hemisphere
- D. Rhombus

464. The principal constituents of the grinding wheel of a grinding machine are made of.....

- A. Stone and Cement
- B. Abrasive Grains and Bond +++
- C. Iron and Stone
- D. Carbide and Bond

465. All the following are the operation that can be carried out on a lathe machine except

- A. Taper Turning +++
- B. Chamfering
- C. Threading
- D. Gear Threading

466. The Controls the depth of cut and is also used for feeding when performing perpendicular, vertical, and angular operations on the shaping machine.

- A. Tool Slide
- B. Clapper Box
- C. Tool Post +++
- D. Cross Post

467. One of these cutting tools is not useful on the milling machine.

- A. End Mill
- B. Gear Cutter
- C. Helical Mill +++
- D. Slab Mill

468. absence of one of these will reduce the efficiency of the cutting tool
- A. Cutting Current +++
 - B. Cutting Voltage
 - C. Cutting Speed
 - D. Cutting Fluid
469. One of the angles on the single point cutting tool used on a lathe machine is called
- A. Clearance angle
 - B. Surface angle +++
 - C. Right angle +++
 - D. Friction angle
470. The name of the cutting fluid derived from crude oil paraffin is -----
- A. Fixed oil
 - B. Soluble oil
 - C. Mineral oil
 - D. Compound oil +++
471.operation is used to create a pattern on the surface of a finished work to increase grip friction
- A. Gear cutting +++
 - B. Knurling
 - C. Drilling
 - D. Threading
472. Drilling is the process of----- while boring is the -----
- A. Originating a non-existing hole; expansion of already drilled hole
 - B. Expanding of already drilled hole; production of a non-existing hole +++
 - C. Originating a large hole; production of small a hole
 - D. Centring an existing hole; development non existing hole
473. Spinning on lathe machine is the----- to a profile/shape
- A. Expansion of metal sheet
 - B. Twisting of metal sheet +++
 - C. Folding of metal sheet
 - D. Drilling of metal sheet

474. and are two types of cutting operation on milling machine

- A. Front and side cut +++
- B. Up and down cut
- C. Up and below cut
- D. Horizontal and vertical cut

475. is the process of removing sharp edge of work pieces on the lathe machine

- A. Conical operation
- B. Cylindrical operation
- C. Round turning operation +++
- D. Chamfering operation

476. Calculate the no of holes to make 8 equal division slots on a 80mm diameter work piece on a milling machine using the Simple indexing method with the following index numbers;17,19,21,24,29,33,43,49

- A. 5
- B. 4
- C. 3 +++
- D. 2

477. One of the purposes of grinding operation is to

- A. Add wanted material
- B. Machine very hard material +++
- C. Produce new holes
- D. Expand already drilled hole

478. High carbon steel tool has % of carbon

- A. 0.83 +++
- B. 0.73
- C. 0.63
- D. 0.50

479. An example of single point cutting tool is.....

- A. Boring tool +++
- B. Milling cutter

- C. Drill cutter
- D. Lathe reamer

480. All are cutting fluid except.....

- A. Mineral oil
- B. Soluble oil
- C. Fixed oil
- D. Variable oil +++

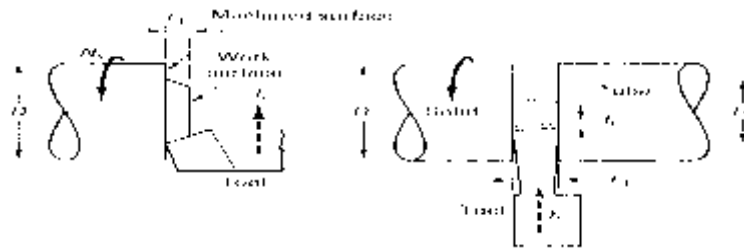
481. All are cutting tool materials except.....

- A. Tungsten carbide
- B. High speed steel
- C. Ceramics
- D. Mild steel +++

482. Screw cutting operation is also called

- A. Knurling operation
- B. Pattern cutting
- C. Filling operation
- D. Thread cutting operation +++

483. The diagram below shows what type of operational sketch on lathe machine



- A. Facing and cut off +++
- B. Taper turning and screw cutting
- C. Threading and reaming operation
- D. Chamfering and boring operation

484. The full meaning of HSS, as a material for cutting tool is

- A. High Steel Stud
- B. High Speed Steel +++
- C. High Steel Speed

D. Horned Speed Steel

485. Cutting fluid perform the following functions except.....

A. Cool the cutting tool and the work piece

B. Reduce friction

C. Increase the cutting angle +++

D. Remove the chips

486. The process of finishing already drilled hole to desirable surface finish is called ...

A. Reaming +++

B. Drilling

C. Centre Boring

D. Counter boring

487. All are examples of protective equipment on the lathe machine except

A. Safety boot

B. Helmet

C. Machine guide +++

D. Eye goggle

488. The part of a Lathe machine which controls the entire operation is called

A. Headstock +++

B. Tailstock

C. Bed

D. Carriage

489. The part of a Lathe machine which create a pathway upon which other parts move or slide is called .

A. Headstock

B. Bed +++

C. Tailstock bed

D. Carriage bed

490. An example of a machine tool in which the cutting tool translates (to and fro) while the work piece is stationary is.

A. Shaping Machine +++

B. Lathe Machine

- C. Drilling Machine
- D. Grinding Machine

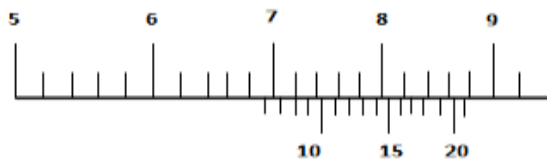
491. The work piece is stationary while the cutting tool rotates in the following machines except

- A. Lathe Machine +++
- B. Drilling Machine
- C. Grinding Machine
- D. Shaping Machine

492. All the following operations can be performed on drilling machine except

- A. Centre Drilling +++
- B. Boring
- C. Semibore drilling
- D. Counter boring

493. What is the reading on the figure shown below:



- A. 6.93 mm
- B. 6.39 cm
- C. 6.93 cm +++
- D. 6.39 mm

494. is the process by which engineers, technicians, craftsmen and every staff in the workshop carry out their duties without sustaining injuries or damaging any equipment

- A. Safety +++
- B. Rules and Regulations
- C. Industrial Act
- D. Work Act

495. Slip gauge are supplied in sets and are produced in one of four grades of accuracy except

- A. Inspection Grade

- B. Reference Grade
- C. Machine Grade +++
- D. Workshop Grade

496. enables us to measure angles to an accuracy of 5 minutes (60 minutes = 1 degree).

- A. Vernier caliper
- B. Vernier bevel protractor +++
- C. Micrometer
- D. Slip Gauge

497. The amount of energy a material can absorb before fracture is

- A. Ductility
- B. True Fracture Strength
- C. Toughness +++
- D. Hardness

498. The measure of how much deformation or strain a material can withstand before breaking is

- A. Elastic Limit
- B. True Fracture Strength
- C. Yield Strength
- D. Ductility +++

499. Fiber-glass is a good example of materials.

- A. Composites +++
- B. Ceramics
- C. Semiconductors
- D. Polymers

500. All the following materials are ceramics except

- A. Alumina
- B. Fiberglass +++
- C. Carbons and graphite's
- D. Silicon carbide

501. Taper turning is made possible on the lathe because the lathe is

- A. Down Cut
- B. Onset
- C. Offset +++
- D. Under cut

502. On the lathe, the cross slide is used to take cut while the compound slide does the feed by hand.

- A. Angular longitudinal cut
- B. Angular parallel cut
- C. Angular horizontal cut
- D. Angular transverse cut +++

503. operation is carried out to enlarge an existing drilled hole while is the process of finishing already drilled hole to desired size.

- A. Boring/Reaming +++
- B. Boring/Drilling
- C. Reaming/Boring
- D. Drilling/Reaming

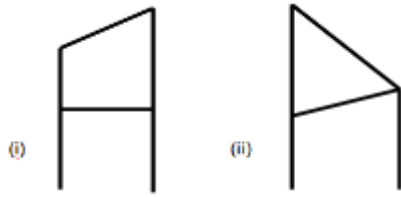
504. These are types of milling cutters except

- A. corner round
- B. round nose +++
- C. convex formed
- D. gear tooth

505. The following are the merits of down-cut milling except

- A. it helps in the cutter's and machine's life span
- B. it gives the work piece a good and smooth finishing surface
- C. a beginner can use it +++
- D. it enables the cutter to bite into the work piece immediately and reduces the load off the tooth gradually

506. Identify the tools below (i) & (ii)



- A. Finishing/Roughing
- B. Facing/Roughing
- C. Facing//Finishing
- D. Roughing/Finishing +++

507. During drilling operations, it is advisable to

- A. fix the drill into the drill chuck properly
- B. not make use of coolant when drilling +++
- C. make sure the drills are running straight before starting the operation
- D. never clean away chips with your hand

508. A fine grained grinding wheel is used to grind hard materials

- A. Incorrect
- B. Correct +++
- C. Correct and incorrect
- D. None of the above

509. is above the carriage

- A. Tail Stock
- B. Tool Post +++
- C. Dead Stock
- D. Compound Slide

510. ----- is reduced in turning

- A. Thickness
- B. Length
- C. Diameter +++
- D. Diameter & Length

511. ----- And ----- threads can be cut on lathe machine

- A. Solid & hollow
- B. Internal & external +++

- C. Upper & down
- D. All of the above

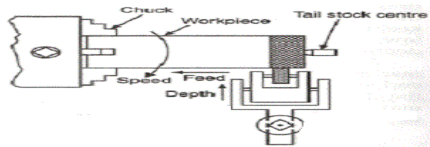
512. ----- is an accessory used on the milling machine in the accurate division of parts

- A. Dividing head +++
- B. Faceplate
- C. Steady rest
- D. Arbor bracket

513. The cutting tool in a milling machine is mounted on-----

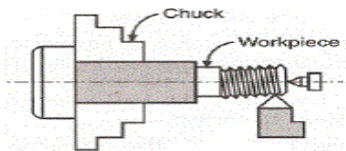
- A. Spindle
- B. Column
- C. Arbor +++
- D. Knee

514. The operational sketch below describe which operation on lathe machine



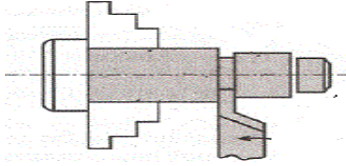
- A. Taper turning
- B. Contour forming
- C. Screw threading
- D. Knurling +++

515. The operational sketch below describe which operation on the lathe machine



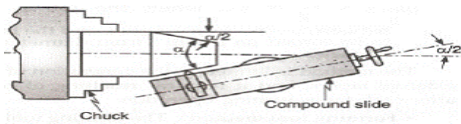
- A. Threading +++
- B. Chamfering
- C. Counter boring
- D. Parting off

516. The sketch below explain which operation on lathe machine



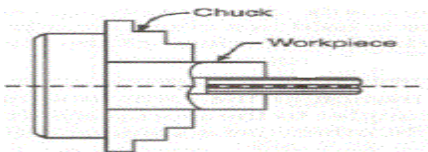
- A. Chamfering
- B. Parting off +++
- C. Slotting
- D. Countersinking

517. The sketch below explain which operation on lathe machine



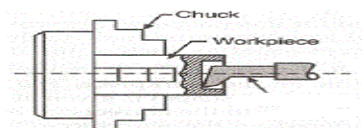
- A. Parallel turning
- B. Step turning
- C. Tapper turning +++
- D. Reaming

518. The sketch below explain which operation on lathe machine



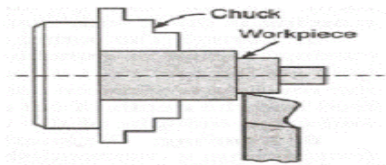
- A. Centre drill
- B. Spot facing
- C. Threading
- D. Reaming +++

519. The sketch below explain which operation on lathe machine



- A. Boring +++
- B. Reaming
- C. Centre drill
- D. All of the above

520. The above sketch explain which operation on lathe machine



- A. Facing turning
- B. Stepping turning +++
- C. Chamfering
- D. Forming

521. All the following cutting can be produced on milling machine except

- A. Gear cutting
- B. Keyways
- C. Chain cutting +++
- D. Slot

522. The followings are the measuring instrument except

- A. Vernier caliper
- B. Micrometer screw gauge
- C. Feeler gauges
- D. Mandrel +++

523. What is the type of operation on lathe machine when the cross slide moves across the bed of the machine

- A. Facing and parting off +++
- B. Taper turning and reaming
- C. Centre drilling and boring
- D. Knurling and counter-boring

524. is the process by which engineers, technicians, craftsmen and every staff in the workshop carry out their duties without sustaining injuries or damaging any equipment

- A. Precautions
- B. Rules and regulations +++
- C. Safety
- D. Work Act

525. Slip gauge are supplied in sets and are produced in one of four grades of accuracy except

- A. Inspection grade
- B. Reference grade
- C. Machine grade +++
- D. Workshop grade

526.enables us to measure angles to an accuracy of 5minutes (60 minutes = 1 degree).

- A. Vernier caliper
- B. Vernier bevel protractor +++
- C. Micrometer
- D. Slip gauge

527. The amount of energy a material can absorb before fracture is

- A. Ductility
- B. True fracture strength
- C. Toughness +++
- D. Hardness

528. The measure of how much deformation or strain a material can withstand before breaking is

- A. Ductility +++
- B. True fracture strength
- C. Yield strength
- D. Elastic limit

529. Taper turning is made possible on the lathe because the lathe is

- A. Off cut
- B. offset +++
- C. Line set
- D. Low cut

530. On the lathe, the cross slide is used to take cut while the compound slide does the feed by hand.

- A. Angular Longitudinal Cut

- B. Angular Parallel Cut
- C. Angular Transverse Cut +++
- D. Angular Horizontal Cut

531. These are types of milling cutters except

- A. Round nose +++
- B. Corner round
- C. Convex formed
- D. Gear tooth

532. The following surface can be produced on a milling machine except

- A. Flat
- B. Involute
- C. Circular +++
- D. Perpendicular

533. Milling machine can be classified according to the position of the -----

- A. Bed
- B. Over arm
- C. Spindle +++
- D. Cutter

534. These are work holding devices on lathe machine except

- A. Vice +++
- B. Mandrel
- C. Fixed steady
- D. Dogger

535. Lathe machine can perform all these function except

- A. Straight turning
- B. Step turning
- C. Drilling
- D. Gear cutting +++

536. The name cross slide is associated with-----

- A. Drilling machine
- B. Lathe machine +++

- C. Milling machine
- D. Grinding machine

537. Coolant is used for -----

- A. Cooling +++
- B. Cutting
- C. Filling
- D. Washing

538. The solution of soluble oil in water is called-----

- A. Milk
- B. Engine oil
- C. Coolant +++
- D. Lubricant

539. Universal milling machine can work as both-----and -----

- A. Plano & vertical miller
- B. CNC & plano miller
- C. CNC & vertical miller
- D. Horizontal & vertical miller +++

540. Knurling operation can be performed on -----

- A. Lathe machine +++
- B. Milling machine
- C. Grinding machine
- D. Drilling machine

541. The name of cutting tools used on milling machine is-----

- A. Cutting tool
- B. Cutters +++
- C. Drill bit
- D. File

542. Grinding machine is used for -----

- A. Slotting
- B. Reaming
- C. Grinding +++

D. Knurling

543. Drill bits are used on this machine

A. Drilling machine +++

B. Milling machine

C. Grinding machine

D. Slotting machine

544. The following are used on milling machine except

A. Cutters

B. Dividing head

C. Vice

D. Punch +++

545. Four jaw independent chuck can be used to hold

A. Circular shape work

B. Irregular shape work +++

C. Cylindrical shape work

D. None of the above

546. The pillar drilling machine is called pillar because it is standing as a

A. Column

B. Bolt

C. Stick +++

D. None of the above

547. Universal scroll chuck can be used to hold-----

A. Circular work +++

B. Irregular work

C. Cylindrical work

D. None of the above

548. The important accessory on milling machine to aid cutting of gears to precision is called

A. Anvil

B. Dividing head +++

C. Divider

D. Cutter

549. When the cutting tool is set too high, the following problems are encountered except

- A. The rake angle increases for external turning
- B. The rake angle decreases for boring
- C. The clearance angle increase for boring
- D. The rake angle increases for boring +++

550. When the cutting tool is set too low, the following problems are encountered except

- A. Tool tends to rub the work
- B. The clearance angle decreases for boring
- C. The tool tends to bite into work during boring +++
- D. The rake angle increases for boring

551. On milling machine, -----can move up and down on the frame or column and it carries the cross slide

- A. The table
- B. The spindle
- C. Knee +++
- D. The over arm

552. Is a device that transforms a physical variable of interest into a form that is suitable for recording.

- A. Measurement
- B. Standard
- C. Instrument +++
- D. Calibration

553. The international standard metre is kept in which country of the world?

- A. London
- B. Zurich
- C. Brussels
- D. Paris +++

554. The type of slip gauge used in the inspection of production parts is called

- A. Calibration grade
 - B. Reference grade
 - C. Inspection grade
 - D. Workshop grade +++
555. The imperial standard yard is held at the board of trade in
- A. London +++
 - B. Zurich
 - C. Brussels
 - D. Paris
556. The Is used for making out vertical and angular dimensions on work piece.
- A. Vee block
 - B. Marking out block
 - C. Scribing block +++
 - D. Slip gauge
557. Is an instrument that cannot be used for direct measurement but must be used to transfer a size to or from a measuring instrument.
- A. Dividers
 - B. Calipers +++
 - C. Dial indicators
 - D. Slip gauge
558. Is the decrease in volume that results from the application of pressure.
- A. Tension
 - B. Strength
 - C. Yield Strength
 - D. Compression +++
559. Is a slowly progressing, permanent deformation that results from a steady force acting on a material.
- A. Toughness
 - B. Ductility
 - C. Creep +++

D. Fatigue

560. Is the rate of heat transfer through a material in a steady state.

A. Thermal diffusivity

B. Thermal conductivity +++

C. Thermal expansion coefficient

D. Thermal resistance

561. The following are the characteristics of Alumina ceramic except

A. Good strength and hardness

B. Low dielectric loss

C. Wear and temperature resistant

D. Used for tool tips +++

562. Slip gauges are supplied in sets and produced in the following grades except

A. Workshop grade

B. Inspection grades

C. Sizing grades +++

D. Reference grades

563. All these are measuring devices commonly used in workshop measurement except

A. Engineer square

B. Precision steel rule

C. Micrometers

D. Compound slide +++

564. is a measuring instrument that is used when a dimension is to be machined within 1mm accuracy.

A. Divider

B. Vernier caliper +++

C. Vee block

D. Slip gauges

565. The following are basic classifications of solid materials except

A. Metals

- B. Crystals +++
- C. Polymers
- D. Ceramics

566. The following are methods of heat treatment except

- A. Softening +++
- B. Normalizing
- C. Annealing
- D. Quenching

SECTION E: WELDING

567. The devices used to ensure uniform accuracy in mass production/fabrication is called.....

- A. Gangs & Steadies
- B. Machine Vice and Clamp holder
- C. Jigs and fixtures +++
- D. All of the above

568. The function of a flux in welding is to prevent.....

- A. Corrosion
- B. Oxidation +++
- C. Heat loss
- D. All of the above

569. What is the minimum distance accepted as arc length in arc welding process.....

- A. 0.30 mm
- B. 20 mm
- C. 50 mm
- D. 3.0 mm +++

570. Cracking is caused by the followings EXPECT

- A. Current too High +++
- B. Rigid Joint
- C. Fast Cooling Rate
- D. Traveling too Slow

571. Which of the following diameter of electrode in millimeter is equivalent to gauge 4?
- A. 0.80
 - B. 8.00 +++
 - C. 0.08
 - D. 0.008
572. The hole on the surface of an anvil which is wider than the punch hole is called thehole.
- A. Clearance
 - B. Punch
 - C. Handle +++
 - D. All of the above
573. How many types of flames do we have in oxy-acetylene welding operation?
- A. 4
 - B. 3 +++
 - C. 2
 - D. 5
574. What type of joint is achieved by welding?
- A. Fixed
 - B. Temporary
 - C. Permanent +++
 - D. Welded
575. What is the full meaning of TIG welding?
- A. Titanium Iodine Gas
 - B. Tantalum Inert Gas
 - C. Tungsten Iodine Gas
 - D. Tungsten Inert Gas +++
576. In TIG welding, what material is the electrode made off?
- A. Copper
 - B. Tungsten +++
 - C. Steel

D. Aluminum

577. The identification of an oxygen filled cylinder is

- A. Green
- B. Yellow
- C. Blue +++
- D. Red

578. The identification color of an acetylene filled cylinder is

- A. Green
- B. Yellow
- C. Blue
- D. Red +++

579. Which of the oxy-acetylene flame is suitable for welding ferrous materials?

- i. Oxidizing
 - ii. Carburizing
 - iii. Neutral
- A. II only
 - B. II or III +++
 - C. III only
 - D. I only

580. Welding shield is used to prevent-----

- A. Blindness
- B. Ray-eye
- C. Shortsightedness
- D. Arc-eye +++

581. Welding can be defined as the process of joining two similar materials under the influence of

- A. Temperature
- B. Pressure
- C. Heat +++
- D. Joint

582. The following are the advantages of rightward welding techniques EXCEPT?

- A. It is safer +++
- B. It is cheaper
- C. Distortion is reduced
- D. It is quicker

583. Which of the following is incorrect about the leftward welding techniques?

- A. The rod leads the flame
- B. The rod is placed at angle between 30° and 40°
- C. The flame is placed at angle between 40° and 50° +++
- D. the flame is placed at angle between 60° and 70°

584. The two terminals of the electric arc machine are ----- and -----

- A. Current and Earth +++
- B. Copper and Bronze
- C. Current and Direct
- D. Copper and Current

585. What is the full meaning of MIG in welding?

- A. Magnesium Iodine Gas
- B. Metallic Inert Gas +++
- C. Manganese Inert Gas
- D. Molecular Inert Gas

586. The followings are classified under chemical source of heat in welding operation chart EXCEPT

- A. Forge
- B. Oxyacetylene
- C. Frictional +++
- D. Argon

587. Spot welding process is classified under source of heat

- A. Chemical
- B. Resistance +++
- C. Frictional
- D. None of the above

588. Seam, Spot and Butt welding processes are similar because heat required for welding is generated by.....

- A. Chemical reaction between a fuel and air in the presence of flame
- B. An electrical energy which is converted to an arc
- C. The resistance to the flow of electric current +++
- D. None of the above.

589. The major constituents of a solder are

- A. Tin and Lead with small Antimony +++
- B. Tin and Lead
- C. Copper and Lead
- D. Copper and Lead with small Antimony

590. is the typical example of a flux used in soldering

- A. Kerosene
- B. Brake Oil
- C. Zinc chloride +++
- D. Acid

591. Brazing is classified as a hard soldering with temperature of about.....

- A. 600°C +++
- B. 660°C
- C. 606°C
- D. 060°C

592. The followings are the function of a regulator except

- A. Regulates the pressure and flow rate of gases required for the welding flame
- B. Determine the gas content +++
- C. Indicates the pressure of gas in the cylinder
- D. Indicates the pressure of gas flowing to the blowpipe

593. How many orifices do an oxyacetylene gas cutting blowpipe has?

- A. 5
- B. 3
- C. 1
- D. 2 +++

594. What type of gas is supplied to the cutting blowpipe when the trigger is depressed during oxy-acetylene flame cutting operation?
- A. None of the above +++
 - B. Acetylene
 - C. Hydrogen
 - D. A mixture of oxygen and acetylene
595. Which of the following gas is suitable for shielding in TIG and MIG welding operation?
- A. Hydrogen
 - B. Carbon dioxide
 - C. Nitrogen
 - D. None of the above +++
596. What type of filler metal is used in MIG welding operation?
- A. Bare metal rods
 - B. Bare metal coils +++
 - C. Coated rods
 - D. All of the above
597. The coated rods are suitable as filler material in one of the welding process which is.....
- A. Frictional welding
 - B. Electric-arc welding +++
 - C. Spot welding
 - D. Forge welding
598. In gas welding process, the function of the spark lighter is
- A. To ignite the oxygen gas
 - B. To ignite nitrogen gas
 - C. To ignite H₂O gas
 - D. To ignite acetylene gas +++
599. Leftward gas welding technique is suitable for welding plates with thickness.....
- A. Greater than 6 mm
 - B. Less than 6 mm +++

- C. Greater than 0.6 mm
- D. Less than 0.6 mm

600. The amperage suitable for gauge 16 or 1.5 mm diameter electrode is

- A. 3.0
- B. 30 +++
- C. 300
- D. 0.30

601. The amperage suitable for gauge 14 or 2 mm diameter electrode is

- A. 50 +++
- B. 55
- C. 60
- D. 65

602. Which of the following diameter of electrode in millimeter is equivalent to gauge 12?

- A. 5.5
- B. 5 +++
- C. 2.5
- D. None of the above

603. In welding, what is the full meaning of AC?

- A. Arc Cracking
- B. Alternating Contact
- C. Alternating Current +++
- D. Arc Contact

604. The device that helps to step up or step down voltage/amperage in electric arc welding is called

- A. Stabilizer
- B. Transformer +++
- C. Inverter
- D. Regulator

605. The electrode holder in electric arc welding process is

- A. Clamp

- B. Vice
- C. Tong +++
- D. Pliers

606. The following except one is not a defect in welding operation

- A. Poor appearance
- B. Cracking
- C. Porosity
- D. Slag +++

607. Which of the following option is the remedy for undercutting defect in welding operation?

- A. Adjust welding current
- B. Reduce the welding speed
- C. Proper holding of the electrode
- D. All of the above +++

608. How many electrode terminals does a spot welding machine have?

- A. 3
- B. 2 +++
- C. 1
- D. 4

609. Bolts and nuts are classified under what type of joint.....

- A. Partial
- B. Permanent
- C. Weak
- D. Temporary +++

610. Borax flux is in which form?

- A. Powder +++
- B. Liquid
- C. Gaseous
- D. None of the above

611. The electrodes of seam welding equipment are in what shape?

- A. Triangular

- B. Conical
- C. Cylindrical +++
- D. Hexagonal

612. Flash welding is similar to butt welding; however heat is generated by between the two ends to be joined

- A. Friction
- B. Arc +++
- C. Ignition
- D. Pressure

613. Which of the following options is not a safe way to keep the gas cylinder in the gas welding workshop?

- A. Keep Greases/Oils always on the cylinders to lubricate them +++
- B. Do not use flame to detect leaks
- C. Cylinder valves must always be closed
- D. Use soap bubbles to detect leaks

614. In the gas welding workshop, the following precautions are taken except

- A. Hands must be protected from heat and metal sparks with shields +++
- B. Protective clothing such as apron and gauntlets are essential to protect the body
- C. Eyes must be protected from glare and splatter by wearing suitable goggles
- D. The equipment should be kept and used in a self-ventilated room or space

615. All of these except is not a safety precaution in the electric arc welding

- A. All equipment must be properly earthed
- B. Keep all the materials to be welded earthed +++
- C. All cables must be properly insulated
- D. Hard boots and protective shoes are essential

616. Which of the following is not among the personal protective equipment peculiar to welding workshop?

- A. Welding Shield
- B. Apron
- C. Earthen Equipment +++

D. Hard boots

617. Which of the following form of welding is the oldest?

A. Spot Welding

B. Forge Welding +++

C. Frictional Welding

D. Seam Welding

618. The following are the functions of a flux coated electrode in arc welding process except

A. Forms Slag

B. Insulates the filler rod

C. Stabilizes the arc

D. Oxidizes the joints +++

619. The form of resistance welding involving the embossing one of the parts to be joined is called

A. Seam Welding

B. Projection Welding +++

C. Flash Welding

D. Spot Welding

620. Which of the following gas welding flames has excess mixture of oxygen than acetylene?

A. Oxidizing flame +++

B. Carburizing flame

C. Neutral flame

D. Blue flame

621. The gas welding flame that has three feathers is.....

A. Carburizing +++

B. Oxidizing

C. Neutral

D. None of the above

622. How many gas welding techniques are commonly accepted?

A. 4

- B. 3
- C. 5
- D. 2 +++

623. One of the following is used for finishing purposes after welding

- A. Hack saw
- B. Hand file +++
- C. Scribe
- D. Punches

624. Which of the following is used to remove slag on an electric arc welded joint?

- A. Mallet hammers
- B. Hack saw
- C. Pliers
- D. Chipping hammer +++

625. Which of the following tools is used for cutting metallic parts?

- A. Scissor
- B. Pliers
- C. Hack saw +++
- D. Hammer

626. Which of the following flames in oxy-acetylene welding is the hottest?

- A. Neutral
- B. Oxidizing +++
- C. Blue
- D. Carburizing

627. Soldering is majorly used for joining which are not subjected to heat and vibrations

- A. Light articles +++
- B. Thick articles
- C. Hard articles
- D. Soft articles

628. The molten metal used in soldering is called.....

- A. Flux

- B. Filler metal
- C. Solder +++
- D. Electrode

629. Which of these is not a permanent joint?

- A. Glues
- B. Adhesives
- C. Welds
- D. Bolts and Nuts +++

630. Which of these is not a chemical welding?

- A. Oxygen Welding +++
- B. Oxy-acetylene Welding
- C. Argon Welding
- D. Forge welding

631. We can achieve two types of joints in joining process, which of these are the two types of joints?

- A. Fixed and removable
- B. Welding and Riveting
- C. Temporary and permanent +++
- D. Forced and Free

632. Which of the following is a type of electrical welding?

- A. Oxy-acetylene welding
- B. Arc Welding +++
- C. Frictional Welding
- D. Gas-shielded welding

633. Forge welding is the oldest form of welding. Why?

- A. It uses forge Machine
- B. It uses fire and metal
- C. The metals are not refined
- D. It uses primitive hand tools +++

634. Another name for resistance welding is.....

- A. Arc Welding

- B. Pressure Welding +++
- C. Electrical Welding
- D. Chemical Welding

635. The type of welding used instead of riveting is.....

- A. Electrical Welding
- B. Arc Welding
- C. Spot Welding +++
- D. Forge Welding

636. The Appropriate type of welding for joining rods, tubes and wires is.....

- A. Flash Welding
- B. Butt Welding +++
- C. Soldering
- D. Seam Welding

637. The following are the examples of fluxes except

- A. Chlorine +++
- B. Resin core
- C. Borax
- D. Tallow

638. The higher reading pressure gauge in oxy-acetylene welding indicates the pressure in.....

- A. Hose
- B. Orifice
- C. Cylinder +++
- D. Nozzle

639. The higher reading pressure gauge in oxy-acetylene welding indicates the pressure in.....

- A. Hose +++
- B. Orifice
- C. Cylinder
- D. Nozzle

640. The cylinders used in oxy-acetylene welding are made to.....

- A. Withstand high pressure +++
- B. Withstand high Temperature
- C. Contain high volume
- D. Resist high heat

641. The presence of inner blue luminous cone in a carburizing flame indicates.....

- A. Presence of Acetylene in the flame
- B. Presence of carbon in the flame
- C. Presence of oxygen in the flame +++
- D. Little acetylene in the flame

642. The carburizing flame is used to add to welded mild steel.

- A. Acetylene
- B. Oxygen
- C. Carbon +++
- D. Nitrogen

643. Which of these metals is inappropriate for bare metal rod?

- A. Aluminium +++
- B. Mild steel
- C. Nickel Steel
- D. Bronze

644. Which of the following is wrong about rightward welding techniques?

- A. Flame leads the rod
- B. rod leads the flame +++
- C. Both rod and flame are at angles between 40° and 50°
- D. Welding proceeds from left to right.

645. Try- square as an ancillary tool in welding is used to ensure

- A. angle 180 degree
- B. angle 90 degree +++
- C. angle 60 degree
- D. angle 30 degree

SECTION F: GENERAL

646. What modification makes spot welding a seam welding?

- A. roller +++
- B. electrode
- C. edge
- D. point

647. The following are ancillary tools used in welding except.....

- A. hack saw
- B. welding torch +++
- C. hammer
- D. hand file

648. One of these is not a form of welding

- A. Mechanical +++
- B. Chemical
- C. Frictional
- D. Electrical

649. Gas-shielded arc welding makes use of gases

- A. Atmospheric oxygen
- B. Nitrogen
- C. Inert
- D. Oxygen and Acetylene +++

650. is the process of uniting two materials

- A. Joining +++
- B. Welding
- C. Forming
- D. Adhesion

651. Which is not a type of pressure welding?

- A. Projection welding
- B. Arc welding +++
- C. Spot welding
- D. Seam welding

652. The amount and type of surface operation depends mainly on all except.....

- A. Types of joint
- B. Material thickness
- C. Welding Process
- D. Length of Material +++

653. The chemical process that occurs between oxygen and acetylene mixture is gas welding is called.....

- A. oxidation
- B. Combustion +++
- C. Burning
- D. D. Heating

654. The head of the cylinder bottle has pressure gauge(s)

- A. 1
- B. 2 +++
- C. 3
- D. 4

655. The bottles used in oxy-acetylene welding can be identified with how many principles

- A. 1
- B. 2 +++
- C. 3
- D. 4

656. is/are used for cylinder bottle identification in oxy-acetylene welding

- A. Colour codes and threading +++
- B. Colour codes only
- C. Threading only
- D. None

657. All are the types of arc welding except

- A. Bare metal
- B. Shielded
- C. Submerged

D. Merged +++

658. Oxygen bottles are having which kind of threading

A. Right hand threading +++

B. Left hand threading

C. Inner threading

D. Outer Threading

659. Acetylene bottles are having which kind of Threading

A. Right hand threading

B. Left hand threading +++

C. Inner threading

D. Outer Threading

660. The principle of flame cutting in gas welding are all except one

A. Sectioning through cutting head

B. Preheating

C. Cutting action

D. Igniting +++

661. The temperature to which the flame from the outer orifice preheats to is

A. 600°C

B. 700°C

C. 900°C +++

D. 800°C

662. Heat for welding purposes can be generated in forms

A. 1

B. 2

C. 3 +++

D. 4

663. Why must the two metals to be welded similar?

A. They must be similar to produce the joint

B. They must melt at the same temperature +++

C. To produce neat welded surface

D. For good surface finish

664. offers greater scope as the heat source for welding

- A. Electrical source +++
- B. Mechanical source
- C. Chemical source
- D. Frictional source

665. Forge welding is achieved by.....

- A. Hammering
- B. Heating
- C. Forging
- D. Heating and Hammering +++

666. When the proportion of acetylene is more than oxygen flame is produced

- A. Carburizing +++
- B. Oxidizing
- C. Neutral
- D. Acetylene

667. When the proportion of oxygen is more than acetylene, flame is produced

- A. Carburizing
- B. Oxidizing +++
- C. Neutral
- D. Acetylene

668. Filler metals are available in how many forms

- A. 1
- B. 2
- C. 3 +++
- D. 4

669. All are forms of filler metals except

- A. Bare metal rod
- B. Bare metal coil
- C. Coated coil +++
- D. Coated rod

670. The filler metal/rod used in electric arc welding is known as

- A. coated rod
- B. Electrode +++
- C. metal coil
- D. brazing iron

671. The coating on the electrode serves the following purposes

- I. Produces a gas shield
- II. Stabilizes the arc
- III. Acts as flux

- A. I only
- B. II and III only
- C. III only
- D. I, II and III +++

672. The distance between the electrode tip and the work-piece is called.....

- A. Arc length +++
- B. Electrode distance
- C. Work-piece distance
- D. tip-piece distance

673. In friction welding, which of the following is correct about the work-pieces?

- A. Both are rotating
- B. One rotating, one stationary +++
- C. Both are stationary
- D. none of the above

674. Which of the oxy-acetylene welding techniques is best suited for thickness less than 6 mm in mild steel plate and non-ferrous metal

- A. Leftward +++
- B. Rightward
- C. either rightward or leftward
- D. neither rightward nor leftward

675. Which of the oxy-acetylene welding techniques is used for steel plates above 6 mm thickness?

- A. Leftward
- B. Rightward +++
- C. either rightward or leftward
- D. neither rightward nor leftward

676. When the proportion of oxygen and acetylene are equal, the kind of flame produced is.....

- A. natural flame
- B. neutral flame +++
- C. oxy-acetylene flame
- D. carburizing flame

677. What could be the implication of absence of oxygen gas during oxy-acetylene welding

- A. No combustion +++
- B. Poor welding
- C. Defect in welding
- D. None of the above

678. The kind of joints produced by glues or adhesives is

- A. Temporary joint
- B. Permanent joint +++
- C. Fixed joint
- D. Removable joint

679. The kind of joints produced by rivets is

- A. welded joint
- B. Permanent joint +++
- C. temporary joint
- D. Removable joint

680. Source of heat in electric arc welding is.....

- A. AC only
- B. DC only
- C. AC and DC
- D. AC or DC +++

681. Which is opened first among oxygen and acetylene control valves during oxy-acetylene welding?
- A. Oxygen
 - B. Acetylene +++
 - C. They are both opened together
 - D. They are not opened together
682. The shield or coating of the electrode is a hardened flux that consists of a mixture of the following except
- A. Cellulose material
 - B. Silica
 - C. Coating manganese +++
 - D. De-oxidants
683. The burning of the cellulose material gives.....
- A. a protective shield of gaseous shield around the weld +++
 - B. slags around the weld
 - C. coatings over the weld
 - D. layers of electrodes over the weld
684. Friction welding occurs as a result of.....
- A. heating to rub
 - B. rubbing to heat +++
 - C. motion
 - D. current
685. The types of joint formed using screws, bolt and nuts are.....
- A. Fixed
 - B. Removable
 - C. Temporary +++
 - D. Permanent
686. The is used to hold the electrode in electric arc welding
- A. Vice
 - B. Tong +++
 - C. Holder

D. Terminal

687. The modification of spot welding that is formed by embossing one of the parts to be joined is called.....

A. seam welding

B. flash welding

C. projection welding +++

D. stitch welding

688. The slip gauges are produced in

A. Grades

B. Reference

C. Slips

D. Sets +++

689. The physical properties of any material range from the following except.....

A. Density

B. Physical Appearances

C. Composites +++

D. Cost

690. The following are examples of ferrous metals except

A. Cast Iron

B. Aluminum +++

C. Tool Steel

D. Stainless Steel

691. is a process of cooling a metal very quickly.

A. Quenching +++

B. Aging

C. Normalizing

D. Annealing

692. The following information should be given when ordering a file.

A. Length

B. Shape

C. Flatness +++

D. Roughness

693. The following are some of the cleaning tools in Bench and fitting practices except

.....

- A. Wire brush
- B. Rigid Scraper
- C. Wire Mesh +++
- D. Wire Wheel

694. The following are examples of cutting fluid in machining operation except

- A. Fixed Oil
- B. Soluble Oil
- C. Mineral Oil
- D. Adsorbent Oil +++

695. The following are the functions of cutting fluids except

- A. Cool the cutting tool and work piece
- B. Remove the Chips
- C. Wash away the Chaffs +++
- D. Provide safe working environment

696. Is the oldest welding process which is accomplished either by hand hammering or by machines.

- A. Pressure Welding
- B. Forge Welding +++
- C. Spot Welding
- D. Scam Welding

697. All the sizes indicated below are the diameter of electrodes except

- A. 1.50mm
- B. 2.00mm
- C. 4.50mm +++
- D. 4.00mm

698. All the following are welding defects except

- A. Poor Appearance
- B. Medium Cutting +++

C. Porosity

D. Brittle Welds

699. The following are ancillary tools for welding operations except

A. Chisel

B. Try-square

C. Tube cutter +++

D. Hacksaw

700. The mixture of two refrigerants to produce R400 series which cannot be separated is called:
- A. Azeotropic
 - B. Zeotropic +++
 - C. Inorganic Refrigerants
 - D. Halogenated Refrigerant
701. Which of the following statement(s) is/are true about the property of a good refrigerant?
- i. Good refrigerants should have low latent heat of vaporization
 - ii. Low C.O.P in the working temperature range
 - iii. Very toxic and flammable
 - iv. It should be able to easily deplete the ozone layer and contribute to global warming
 - v. It should have low specific heat and considerably high thermal conductivity
- A. All of the above
 - B. i, ii, and iv
 - C. iii and iv
 - D. v only +++
702. The winter type of air-conditioning systems is not an application of refrigeration system because
- A. Air is cooled and dehumidified
 - B. Air is heated and dehumidified
 - C. Air is heated and humidified +++
 - D. Air is both cooled and heated
703. The name given to the heat exchanger which ejects heat from the hot refrigerant vapour to the sink thereby changing its phase back to liquid is called:
- A. Evaporator
 - B. Expander
 - C. Refrigerant
 - D. Condenser +++

704. The main process whereby a lower temperature than that of the immediate surrounding can be achieved is called:
- A. Dehumidification
 - B. Adiabatic process
 - C. Refrigeration +++
 - D. Evaporation
705. The treatment of air so as to simultaneously control its temperature, moisture content, motion and distribution for the purpose of comfort requirement as required by the occupants and products or perhaps for industrial means can be specifically classified under:
- A. Refrigeration
 - B. Air-Conditioning +++
 - C. Isothermal Air Treatment
 - D. Evaporation
706. The Evaporator and the inlet side of the compressor in the refrigerating systems fall under which of these regions:
- A. High pressure and temperature region
 - B. Medium pressure region
 - C. Medium temperature region
 - D. Low pressure and temperature region +++
707. Which of the following is not a typical part of a hammer
- A. Wedges
 - B. Shoulder +++
 - C. Eye
 - D. Face
708. The removal of thick layer of metal is known as
- A. Pinning
 - B. Shaping
 - C. Threading
 - D. Chipping +++
709. Parts of typical hacksaw include the following EXCEPT.....

- A. Tang +++
- B. Frame
- C. Wing nut
- D. Hacksaw blade

710. In an electrical circuit, the red yellow/green wire indicates

- A. Neutral
- B. Earth +++
- C. Negative
- D. Positive

711. A device that controls the off and on in an electrical system is called

- A. Switch
- B. Control
- C. Thermostat +++
- D. Thermometer

712. This cannot be a work holding device on the Lathe Machine

- A. Dog
- B. Face plate
- C. Collet +++
- D. Centers

713. This work holding device is usually attached to the lathe spindle so as to hold flat work pieces when it is necessary to drill holes on flat wide thick plates on the lathe machine.

- A. Hinge plate
- B. Face plate +++
- C. Self-driven center
- D. Fixed steady

714. The work piece is fixed to theon the lathe machine, while it is fixed to the table on the milling machine

- A. Spindle +++
- B. Tail stock
- C. Table

- D. Tool post
- 715. All except this are work holding devices on the lathe machine
 - A. Drill chuck
 - B. Self-centering chuck
 - C. Independent chuck
 - D. Center chuck +++
- 716. All these cutting tools can be used on the drilling machine except
 - A. Parallel shank drill
 - B. Tapered shank drill
 - C. Reamer
 - D. Boring drill +++
- 717. Machine tools are usually
 - A. Power driven +++
 - B. Motor driven
 - C. Switch driven
 - D. Operator driven
- 718. One of these operations is only applicable on the milling machine
 - A. Gear cutting +++
 - B. Facing
 - C. Drilling
 - D. Boring operation
- 719. This essential part is absent on the shaping machine
 - A. Spindle +++
 - B. Table vice
 - C. Electric motor
 - D. Adjuster
- 720. The device attached to the milling machine which provides the means of obtaining accurate circular divisions of gear teeth, slots, spline etc is called
 - A. Formula probe
 - B. Divider
 - C. Index formulator

- D. Dividing head +++
721. What will be the hole circle to be used if seven (7) equally spaced divisions is required on the index with the following hole numbers 17,19, 21, 24, 29, 33
- A. 29
 - B. 19
 - C. 33
 - D. 21 +++
722. This is not a type of center on the lathe machine
- A. Pipe centre
 - B. Male center
 - C. V- Centre
 - D. Point center +++
723. Generally, on the lathe machine, work pieces are held between
- A. Holders
 - B. Chucks
 - C. Centers +++
 - D. Carriage
724. Which of the following cutting fluids requires experience in its usage?
- A. Fixed oil
 - B. Paraffin +++
 - C. Mineral oil
 - D. Soluble oil
725. It is always advisable to initiate drilling process by using this tool on the lathe machine
- A. Combination drill +++
 - B. Centre punch
 - C. Twist drill
 - D. Parallel drill
726. The motion of the shaping machine during machining operation
- A. Linear motion
 - B. Translational motion +++

- C. Movement motion
 - D. Still motion
727. One of the following is a single point cutting tool
- A. Flat drill +++
 - B. Twist drill
 - C. Reamer drill
 - D. Counterbore drill
728. The following is a protective tool on the pedestrian grinding machine
- A. Safety boot
 - B. Magnetic glove
 - C. Safety goggle +++
 - D. Helmet
729. One of the following is not a part of shaping machine
- A. Tool post
 - B. Clapper box
 - C. Ram slide way
 - D. Compound slide +++
730. Safety precaution in the workshop is essential so as to avoid
- A. Punishment
 - B. Success
 - C. Loss of mark
 - D. Accident +++
731. A book that contains all essential set of guide and information on the operation of machine tools is generally referred to as
- A. Information manual
 - B. Safety manual
 - C. Maintenance manual
 - D. Instruction manual +++
732. ----- is widely used in residence, offices, commercial buildings, airports, hospitals etc
- A. Refrigeration

- B. Air conditioning
 - C. Refrigeration and air conditioning
 - D. Comfort air conditioning +++
733. Natural means of refrigeration is achieved by the use of
- A. Evaporative cooling process +++
 - B. Condensing cooling process
 - C. Atmospheric cooling process
 - D. Conventional cooling process
734. The cooling of a cup of tea is a ----- process
- A. Refrigeration
 - B. Air conditioning
 - C. Circulation
 - D. Spontaneous cooling +++
735. Refrigeration and cooling imply decrease in temperature
- A. True +++
 - B. Maybe
 - C. False
 - D. I don't know
736. Selection of refrigerant is based on the following properties except
- A. Thermodynamic
 - B. Safety
 - C. Thermo- chemical +++
 - D. Economic
737. ----- are substances used in refrigerating systems as heat carriers.
- A. Fluids
 - B. Evaporators
 - C. Refrigerants +++
 - D. All of the above
738. The following are thermodynamic properties of a refrigerant except -----
- A. Boiling point
 - B. Global warming potential +++

- C. Molecular weight
 - D. Critical temperature
739. Azeotropic mixtures are designated by
- A. 400
 - B. 600
 - C. 500 +++
 - D. 200
740. Types of refrigerants include----- except
- A. Inorganic
 - B. Saturated compounds +++
 - C. Hydrocarbons
 - D. Azeotropic mixtures
741. Application of refrigeration and air-conditioning can be found in so many aspects of life such as
- A. Commercial
 - B. Pharmaceutical +++
 - C. Household
 - D. Mobile
742. Refrigeration involves processes such as
- A. Compression
 - B. Expansion
 - C. Humidification +++
 - D. Cooling
743. Heat rejection takes place in the -----
- A. Compressor
 - B. Evaporator
 - C. Expansion device
 - D. Condenser +++
744. Which of the components is the heat source in a refrigerating system?
- A. Compressor
 - B. Evaporator +++

- C. Expansion device
 - D. Condenser
745. The art of refrigeration can be achieved through the following system except
- A. Vortex or pulse
 - B. Air-Pulse refrigeration +++
 - C. Vapour jet refrigeration
 - D. Vapour condensation refrigeration
746. A good refrigerant must have of the following except
- A. Latent heat of vaporization
 - B. Ozone depletion potential +++
 - C. Chemical stability
 - D. Viscosity
747. The most important component of a VCRS is -----
- A. Compressor +++
 - B. Evaporator
 - C. Expansion device
 - D. Condenser
748. ----- is an example of heat exchanger
- A. Evaporator
 - B. Expansion device
 - C. Condenser +++
 - D. Refrigerant
749. is one of the Major application of refrigeration.
- A. Domestic +++
 - B. Food
 - C. Automobiles
 - D. Air-conditioning
750. All the following tools are essential in the servicing of refrigeration and air-conditioning except-----
- A. Manifold gauge
 - B. Vacuum pump

C. Flaring tool

D. Machine snips +++